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Regional scale simulations of daily suspended sediment concentration at gauged and ungauged rivers using deep learning

Abhinav Gupta *, Dongmei Feng

Department of Chemical and Environmental Engineering, University of Cincinnati, OH, USA

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ABSTRACT

This study examined the simulations of riverine suspended sediment concentration (SSC) using a deep learning model called Long Short-Term Memory Network (LSTM) and areal averages of daily scale meteorological data and watershed attributes as predictor variables. Observed streamflow, although strongly correlated with SSC, was not used as a predictor since it is not measured for most rivers. The SSC data between 1983 and 2017 from 167 USGS stations in the Mississippi River Basin were used. The LSTM models were trained using two objective functions: normalized mean square errors (NMSE) and normalized mean absolute error (NMAE). The models were tested in gauged and hypothetical ungauged scenarios. In the gauged scenario, the LSTM was trained on the data from all the watersheds; in the ungauged scenario, the LSTM was trained using a leave-location-and-time-out scheme. About 78% and 50% of the watersheds had positive Nash-Sutcliffe Efficiencies (NSEs) in the gauged and ungauged scenarios, respectively. A slight improvement in model performance was achieved when streamflow simulated by the variable infiltration capacity model was used as one of the predictors. The model trained on the NMAE instead of the NMSE objective function improved the simulations of low SSCs in several watersheds. The model performance was poor in some of the watersheds due to the small amount of training data available in these watersheds, suggesting that it is crucial to collect riverine SSC data to develop reliable models. Overall, these results are encouraging as the SSC could be simulated accurately in many of the watersheds.

1. Introduction

Information on riverine suspended sediment concentration (SSC) is required for a comprehensive ecological assessment and various engineering projects. However, in-situ observations of SSC are sparse both in time and space (Zhi et al., 2024). Therefore, different types of mathematical models, including process-based (PB) models (e.g., Himanshu et al., 2017), simple statistical models (e.g., Runkel et al., 2004), and machine learning (ML) models (Rajaee and Jafari, 2020), have been developed to simulate and forecast SSC.

The PB and ML models have their advantages and disadvantages. PB models are based on knowledge about the processes governing the SSC dynamics within a watershed and, therefore, have a strong inductive bias (Mitchell, 1980). (The term 'inductive bias' refers to all the assumptions behind a model such as mass balance). The knowledge of the SSC dynamics can be useful, especially in data-scarce and ungauged watersheds. However, this knowledge cannot be easily converted to mathematical equations, and often, simpler models must be used to simulate the quantity of interest (Beven, 2011). These models have some effective parameters that cannot be directly measured in the field and must be calibrated against available data (Beven, 2011). The quality of the available data is an important issue in this respect (e.g., Beck, 1987; McMillan et al., 2012, 2018; Gong et al., 2013; Beven, 2011). SSC measurements are expected to be fraught with uncertainties, including measurement and incommensurability uncertainties (Beven, 2011). Precipitation and streamflow data also incur various uncertainties (e.g., Gupta et al., 2023, 2024). Therefore, the calibration of a PB model is a difficult task.

In contrast to the PB models, the ML models have weaker inductive bias and can identify the relationship between the predictor and the response variables based on available data. However, this relationship is often valid only in those regions of the predictor space where some data were available to train the model. Thus, an ML model may require a large amount of training data depending on the complexity of the SSC dynamics in a watershed. Also, the data quality may significantly impact the ML model's capability of finding a good relationship between predictor variables and the SSC. Further, the ML models do not provide much insight into the physical processes because the relationships

* Corresponding author at: 2660 Clifton Ave, Cincinnati, OH, 45221, USA.
E-mail address: gupta4ab@ucmail.uc.edu (A. Gupta).

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identified by the neural networks are compositions of several non-linear functions which are difficult to comprehend.

In the process-based paradigm, the SSC can be simulated by coupling a sediment yield (SY) model with a river routing model. Several empirical models have been used to simulate suspended sediment at long timescales (e.g., yearly), such as the Water Erosion Prediction Project (WEPP, Foster et al., 1987), Universal Soil Loss Equation (USLE, Wischmeier and Smith, 1978), Revised USLE (RUSLE, Renard, 1997), Morgan-Moragn Finney (MMF, Morgan and Duzant, 2008), and others (Patil et al., 2012; Pelletier, 2012). However, routing of sediment to a watershed's outlet is challenging because of the complex effect of river bathymetry, riparian vegetation, and streamflow on the river deposition–erosion dynamics. Tan et al. (2018) compared eight empirical SY models in 450 headwater catchments with areas ranging from 0.1 to 200 km^2. The models were typically poor in reproducing the observed annual average SY (coefficient of determination, $R^2 < 0.20$ for most of the models) for both the calibration and validation catchments, except the MMF model which gave an R^2 of 0.525. Further, Tan et al. (2018) reported large biases in daily SY simulations. Cohen et al. (2014) used a modified Water Balance Model and Sediment (WBMsed) model to simulate sediment fluxes at different sites across the globe, but sediment fluxes were poorly simulated even at the monthly scale. Li et al. (2022) combined the hillslope MMF model and models for riverine and reservoir sediment processes to compute monthly sediment discharges in 35 stations located mostly in the Mississippi River Basin. They found that only 17 out of 35 stations had KGE greater than zero. Although this model was designed to simulate sediment *discharge* at a large scale, the results highlight the difficulty in dealing with various sediment processes in a watershed. Simpler statistical equations such as LOADEST (Load estimator) that relate sediment load to observed streamflow and day of the year have also been explored in the past with mixed success (Runkel et al., 2004; Mallya et al., 2020). These results suggest that while the PB models have some value in reproducing SY, the simulation of SY and riverine SSC at short and long timescales and at large spatial scales (such as catchment scale) remains difficult.

ML models have also been used to simulate and forecast SSC (Kisi et al., 2008; Kisi, 2012; Sirabahenda et al., 2017; Zhu et al., 2007; Cigizoglu, 2004; Lafdani et al., 2013; Rajaee and Jafari, 2020; Salih et al., 2020; AlDahoul et al., 2021). Most of the ML studies used a combination of current and past sequences of rainfall and streamflow to predict SSC using various algorithms, including multiple linear regression (MLR), support vector regression (SVR), and artificial neural networks (ANN). Many studies used past sequences of observed SSC values, effectively forecasting instead of simulating the SSC values. The lengths of training data varied from 50 to 10,000 days in these studies. Liu et al. (2019) reported a Nash-Sutcliffe Efficiency (NSE) of 0.64 for daily SSC prediction using rainfall, streamflow, and NDVI data with ANN. Kişi (2009) predicted daily SSC with R^2 of 0.88 and 0.95 for the Rio Valenciano (drainage area 43.57km^2) and the Quebrada Blanca stations in Puerto Rico (drainage area 8.63km^2), using observed streamflow as a predictor and 1 year of training data. Malik et al. (2017) predicted daily SSC using two algorithms with NSE of 0.97 (co-active neuro-fuzzy inference system) and NSE of 0.82 (MLR), using observed streamflow and observed SSC at previous the timesteps as the predictor variables and 3 years of training data. Jain (2012) predicted daily SSC at two stations with NSE of 0.99 and 0.87 using observed river stage, streamflow, and SSC at the previous timestep with SVR.

The ML studies have shown that streamflow and SSC are often strongly correlated, which has been utilized to develop a sediment rating curve (SRC) (e.g., Walling, 1977; Ferguson, 1986; Crawford, 1991; Asselman, 2000; Warrick, 2015; Zounemat-Kermani et al., 2016). SRC is an empirical power law relationship between SSC and streamflow used to estimate SSC using observed Q. However, the SRC may not always be accurate (Sadeghi and Saeidi, 2010). Further, most watersheds in the world do not have any streamflow measurements (Wagener and Montanari, 2011), and the SRC approach does not account for hysteresis in the SSC dynamics (Smith and Dragovich, 2009; Gao and Josefson, 2012; Fan et al., 2013; Yeshaneh et al., 2014). Therefore, an important question is whether we can simulate SSC using available meteorological data, static watershed attributes, and streamflow simulated by a regional/global hydrological model? The simulated streamflow data from regional hydrological models are available in many parts of the world. For example, simulations of 3-hourly runoff and daily streamflow simulated from the Variable Infiltration Capacity (VIC; Liang, 1994) model are available for the period 1980–2019 across the globe (Yang et al., 2021).

An alternative to hydrometeorological data is remote sensing (RS) data, which have been used to estimate the riverine SSC across the globe or a continent (Dethier et al., 2020; Gardner et al., 2023). However, the RS approach only applies to relatively large rivers determined by the spatial resolution of the RS product. The LANDSAT data frequently used in the RS studies has a revisit period of 8–16 days, resulting in only 3–4 estimates of SSC within a month, depending upon the cloud cover (Friedmann et al., 2024). Thus, the historically available RS data can only yield sparse estimates of SSC.

This study aims to comprehensively assess the predictability of SSC using a deep learning (DL) model called Long Short-Term Memory Network (LSTM) and using hydrometeorological data as input. LSTM has been recently used to predict streamflow in various watersheds around the world and has shown promise as a benchmark model (e.g., Kratzert et al., 2019; Lees et al., 2021). Many studies have used DL models to simulate or forecast water quality parameters (Rahmani et al., 2021; Zhi et al., 2021; Jamei et al., 2023). AlDahoul et al. (2021) used LSTM to predict daily suspended sediment load in the Johor River Basin, Malaysia (drainage area 2, 286km^2) using observed streamflow as the predictor variable and reported an R^2 of 0.92. Similar results have been reported by Pham Van et al. (2023) for a river basin in Vietnam. Song et al. (2024) developed an LSTM for 377 watersheds across the USA using observed streamflow as one of the predictor variables. The LSTM has also been used to *forecast* suspended sediment load (Sahoo et al., 2023). To our knowledge, a comprehensive assessment of LSTM models trained to predict SSC without using observed streamflow as a predictor is missing.

One of the advantages of the LSTM is that it can be trained to extract useful information from multiple watersheds, as opposed to the PB models, which must be tailored to a specific watershed for good performance (Beven, 2000; Nearing et al., 2021). The LSTM model can be trained on hundreds of watersheds to obtain a 'global' model, or it may be tailored to a single watershed to obtain a 'local' model. Previous studies showed that global models typically outperform local models for streamflow predictions (Kratzert et al., 2018, 2024; Nearing et al., 2021; Gupta, 2024) and SSC predictions (Song et al., 2024). Further, the LSTM model can account for hysteresis by efficiently processing the past meteorological inputs into a watershed.

The LSTM can provide continuous estimates of SSC because historical daily meteorological data are readily available for most of the watersheds across the globe. Another advantage of LSTM is that it is relatively easily scalable to simulate daily SSC for the global or continental domain. This study assesses the LSTM's performance in simulating SSC in local, global, and ungauged scenarios. Further, the estimated parameters and model performance may strongly depend upon the loss function considered (Hastie et al., 2009; Kratzert et al., 2019; Gupta and McKenna, 2025). This study analyzed the LSTM models trained using the two popular objective functions – normalized mean absolute error (NMAE) and normalized mean square error (NMSE), which emphasize different parts of the SSC time series. Another question is whether the empirical models typically used to simulate SY have any value in addition to what LSTM trained on meteorological data provides. To answer this question, SY simulated by the MMF model was used as one of the predictor variables.

The objectives of this study were as follows:

(1) To assess the predictability of daily SSC by using the LSTM model in gauged and hypothetical ungauged watersheds where both observed streamflow and SSC data are unavailable,
(2) To analyze the differences in performance of the LSTM models trained using NMSE and NMAE as the loss functions,
(3) To assess the usefulness of VIC simulated streamflow as a predictor in the LSTM model to predict daily SSC,
(4) To assess the usefulness of the MMF model simulated SY as a predictor in the LSTM model to predict daily SSC.

2. Study area and data

The data from the Mississippi River Basin (MRB), the largest river basin in the United States (U.S.) (Fig. 1), were used. The MRB is composed of six subbasins (shown as a subfigure in Fig. 1a): Missouri River Basin (MB), Upper Mississippi River Basin (UMRB), Lower Mississippi River Basin (LMRB), Ohio River Basin (ORB), Tennessee region (TR), and Arkansas Red White region (ARW). The study period was calendar years 1983–2017. The MERIT BASIN (Yamazaki et al., 2017, 2019; Lin et al., 2019) dataset was used for river network and catchment boundary information. The total drainage area of each river could be determined using the river network and corresponding catchment datasets.

SSC data were collected from 250 USGS stations located within the MRB. Some USGS stations were removed from the analysis because they did not intersect with any of the rivers in the MERIT network. The drainage areas of the river were available from two sources: Reported by USGS and calculated using the MERIT dataset. The stations for which the areas obtained by the two sources differed by more than 15 % were excluded to reduce th effect of incommensurability errors. Some river reaches had more than one USGS stations at different locations along the reach; in these cases, the data from the most downstream station were used for the modeling because areal average meteorological and static attribute data (as discussed below) better represent the downstream station. Further, the USGS stations with less than 100 daily observations were excluded. This filtering process resulted in 167 USGS stations, data from which were used for LSTM model development (see Fig. 1 for the locations of these stations).

Fig. 1b shows that most USGS stations had less than 10 years ($\approx$ 3650 days) of daily SSC observations and the watersheds in the western MRB had fewer observations. There were 450,772 daily SSC observations across the 167 watersheds (an average of 2,700 per watershed). The mean SSC ranged between < 10 and 10000mg L^{-1}, and the western MRB recorded much higher SSCs (Fig. 1c, 1d, 1e). Fig. 1f shows that the number of observations was highest in 1986, and a large number of the watersheds had SSC observations before the year 1995, after which the number of watersheds with any SSC observations in a year has been declining.

Several dynamic and static attributes were collected for the drainage areas of the 167 USGS stations (Table 1). Daily Precipitation, daily minimum and maximum temperatures, and daily solar radiation data were obtained from DAYMET (Thornton et al., 2022), available at 1km $\times$ 1km resolution. Additionally, 3-hourly precipitation data from Multi-Source Weighted-Ensemble Precipitation (MSWEP; 0.1° spatial resolution) were used to evaluate the effect of the precipitation products on the modeling results (Beck et al., 2019); 3-hourly data from MSWEP were converted to daily data by summing the 3-hourly data over the day. The gridded meteorological data from the DAYMET and MSWEP were converted to area-weighted averages over the watersheds by first computing the areas of the grids overlapping with a watershed's drainage area and then by computing the areal average of the meteorological variables during a timestep (daily for DAYMET data and 3-hourly for MSWEP data):

$$\overline{x}_t = \frac{\sum_{i=1}^{n} A_i x_{i,t}}{\sum_{i=1}^{n} A_i}, \quad (1)$$

where A_i denotes the overlapping area of the i^{th} grid with a watershed, $x_{i,t}$ denotes the value of a meteorological variable in the i^{th} grid at a timestep t, and $\overline{x}_t$ denotes the areal average of the meteorological variable at the timestep t. The watershed boundaries were obtained from the MERIT BASIN dataset (Lin et al., 2019).

Simulated streamflow values were obtained by routing 3-hourly runoff simulated using the VIC model (Yang et al., 2021) through the river network using a hillslope river routing model (Beighley et al., 2009). Streamflow was scaled by the corresponding drainage area before being used for LSTM modeling. Note that the VIC model was used without any calibration. Many calibrated conceptual hydrological models will produce better streamflow simulations than the uncalibrated VIC model. But these models require streamflow data for calibration, and this study aimed to assess the SSC simulation accuracy in the absence of streamflow data. Therefore, the VIC model is a suitable candidate for this study. To understand the errors introduced by using the VIC simulated streamflow, separate LSTM models were also trained using the observed streamflow. These results, however, are only sparingly discussed in the main text.

We also simulated sediment yield (SY) using the Morgan-Morgan Finney (MMF; Morgan and Duzant, 2008) model, which was used as one of the inputs to the LSTM model. MMF model is an empirical model used to simulate SY at annual timescales (Tan et al., 2018) but has also been used to simulate SY at monthly and daily timescales (Li et al., 2022). In this study, the model was used to simulate daily SY with the expectation that the information codified in the model may be useful for SSC simulated by the LSTM model. The details of the MMF model implementation are provided by Tan et al. (2018).

Monthly Leaf Area Index (LAI) data were obtained from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS, Level 4 Version 6.1, and 500 m spatial resolution). LAI data were downloaded using Google Earth Engine, which combined the data from MODIS Terra (Myneni et al., 2021a) and MODIS Aqua (Myneni et al., 2021b). These LAI data were available after 2002; therefore, average LAI values were used for the missing months. This approximation is reasonable since the *inter-annual* variation of LAI was much smaller than the *intra-annual* variation. The inter-annual variation of LAI in a watershed was computed as the standard deviation of the annual average LAI. The intra-annual variation was computed as the maximum of 'the standard deviation of the monthly LAI within a year' over all the years for which the LAI data were available. The median value of inter-annual and intra-annual variations over all the 167 watersheds were 0.05 and 0.95, respectively, illustrating that the intra-annual variation was significantly larger than the inter-annual variation of LAI.

Sand, silt, and clay content data were collected from the World Data Center-Soil (WDS-Soil, Batjes, 2016), which contains various soil attributes at 30-arcsec spatial resolution. To compute the soil texture in a watershed, WDS-Soil pixels intersecting with a watershed's drainage area were identified, and subsequently, watershed-scale sand, silt, and clay content were computed as means of sand, silt, and clay contents in each of these overlapping pixels. Plane slope was calculated using the MERIT DEM dataset (Yamazaki et al., 2019). The channel length of each river from the MERIT Basin dataset was used to calculate the total channel length within the watershed. The Global Reservoir and Dam dataset (GRanD; Lehner et al., 2011) was used to identify any reservoir within the drainage area of the watersheds. The three reservoir variables, including the total surface area, the total storage capacity of the reservoirs within a watershed, and the average ratio of reservoir storage to distance from the watershed outlet, were used as the predictor variables to represent dam trapping effects. Median bed-material sediment particle size (D50; Abeshu et al., 2021) was also used as a predictor variable. Finally, data on fractions of barren, developed, forest, and

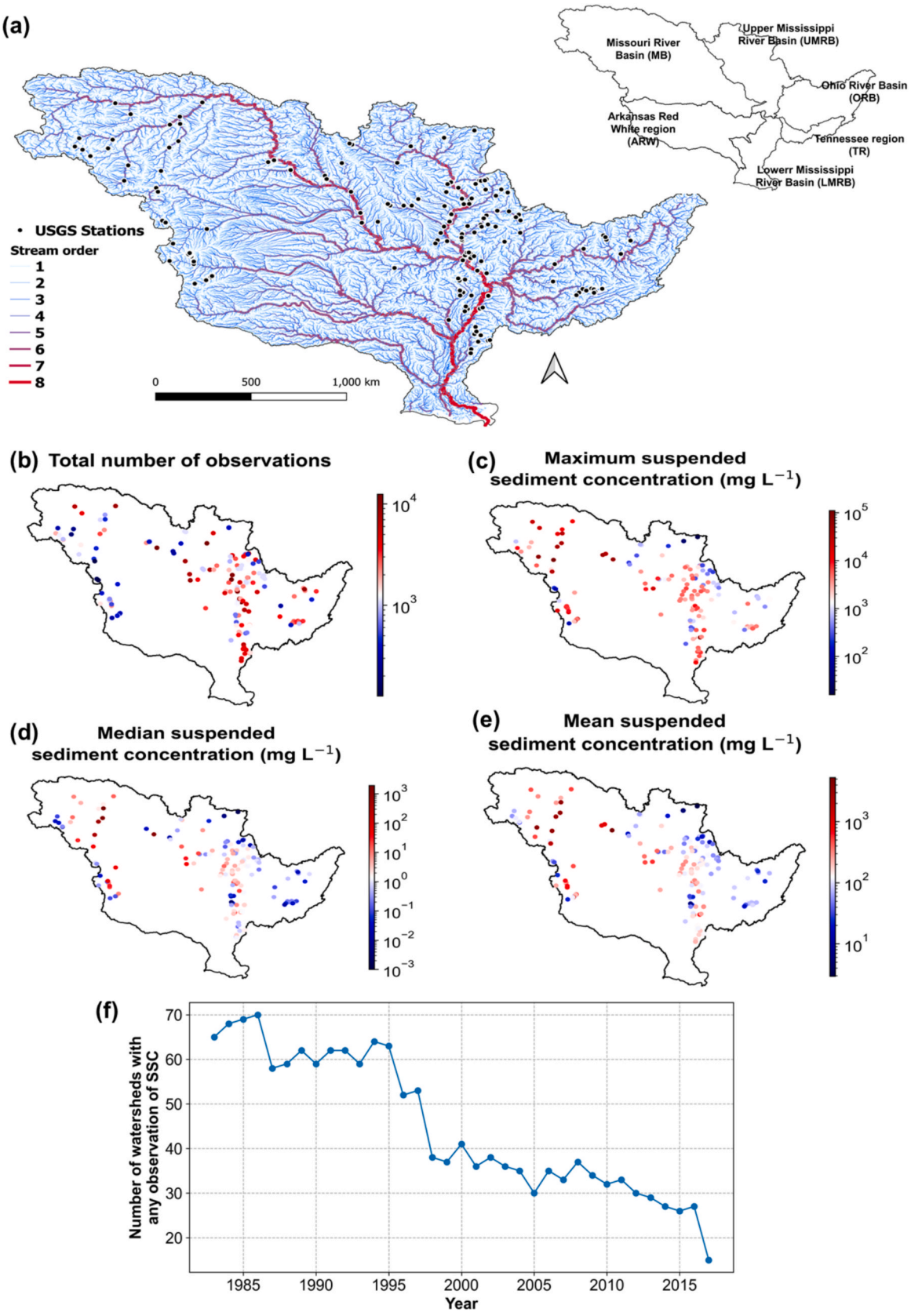

Fig. 1. (a) The locations of SSC stations in the Mississippi river basin used in this study (the river network is mapped based on MERIT BASIN dataset). The red lines are the rivers with larger stream order. The inset map shows different hydrologic region of the MRB. (b) Total number of observations, (c) maximum SSC, (d) median SSC, (e) mean observed SSC in the study watersheds. (f) Number of watersheds that contain at least one SSC observation from a year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Dynamic and static attributes used in the study.

Attributes	Source	Reference
Dynamic		
Daily precipitation	DAYMET and MSWEP	Thornton et al. (2022); Beck et al. (2019)
Daily minimum temperature	DAYMET	Thornton et al. (2022);
Daily maximum temperature	DAYMET	Thornton et al. (2022);
Daily solar radiation	DAYMET	Thornton et al. (2022);
Simulated streamflow	VIC runoff routed using a hillslope routing model	This study
Simulated sediment yield	MMF model	This study
Monthly leaf area index (LAI)	MODIS satellite product	Myneni et al. (2021a, 2021b)
Static		
Sand, silt, and clay content	WDC-Soils	Batjes (2016)
Plane slope	MERIT BASIN	Feng and Gleason (2024)
Total channel length	MERIT BASIN	Lin et al, 2019
Total drainage area	MERIT BASIN	Lin et al, 2019
Total reservoir surface area	GRanD	Lehner et al. (2011);
Total reservoir capacity	GRanD	Lehner et al. (2011);
Average reservoir capacity divided by the distance from the outlet	Derived using the GRanD data	This study
Median bed-material sediment particle size (D50)	–	Abeshu et al. (2021)
Fraction of barren land	NLCD	Dewitz et al. (2021); Wickham et al. (2021); Homer et al. (2020); Jin et al. (2019); Yang et al. (2018)
Fraction of developed land	NLCD	–
Fraction of forest land	NLCD	–
Fraction of shrub land	NLCD	–

VIC: Variable Infiltration Capacity model.
MMF: Morgan-Morgan Finney model.
NLCD: National Land Cover Database.
GRanD: Global Reservoir and Dam Database.
MSWEP: Multi-Source Weighted-Ensemble Precipitation
DAYMET: Daily Surface Weather and Climate Summaries.

shrubland were obtained from the National Land Cover Database (Dewitz, 2021; Wickham et al., 2021; Homer et al., 2020; Jin et al., 2019; Yang et al., 2018). The NLCD provides pixel-level landcover data, which was converted to watershed scale by first identifying the pixels intersecting with a watershed's drainage area and then computing the fractions of these pixels labeled as barren, developed, forest, and shrubland. The land cover data are available for several years at 2–3 years of gap. In this study, the data for the year 2001 were used as it falls in the middle of the study period. The maps showing the spatial distribution of various dynamic and static attributes have been provided in Appendix A (Figs. A1-A3).

3. Model development

3.1. Experimental setup

The LSTM is a variant of the recurrent neural networks, especially suitable for time series predictions. Detailed descriptions of the recurrent neural networks and the LSTM can be found in Goodfellow et al. (2016) and Hochreiter and Schmidhuber (1997). The LSTM has the advantage that it can efficiently process a long sequence of past meteorological variables to simulate the SSC at the current timestep.

The data from each watershed were divided into training and testing periods. The period containing the first 60 % of the days with SSC measurements was used as the training period, and the period containing the remaining 40 % of the days with SSC data was used as the testing period. For example, if a watershed had 200 daily observations, the first 120 observations in the time sequence were used to train the models, and the last 80 observations were used to test the model performance. Since the numbers and periods of days with SSC measurements varied across the watersheds, the training and testing sets had different numbers and periods of observations from different watersheds. Note that the temporal splitting of the time series data into training and testing periods mimics the real-world applications of environmental models where historical data are used to calibrate/train a model, which is then used to simulate the quantity of interest in some future period or hypothetical changed conditions. Ideally, the data should be divided into training, validation, and testing sets, where the validation data should be used for hyperparameter tuning. However, several of the 167 watersheds had limited SSC observations, which inhibited dividing the data from a watershed into the three sets.

Three types of LSTM models were developed to test the SSC predictability in gauged and ungauged scenarios:

(1) Local models: These models were tailored to a specific watershed. These models were trained on data from a single watershed and were used to predict SSC in the testing phase in the same watershed. Watershed static attributes were not used as inputs in these models because the static attributes corresponding to a watershed were assumed to be constant. A total of 167 local models were trained, one for each of the 167 watersheds for a given set of predictor variables.
(2) Global model: A single model was trained using training data from all the watersheds, which was then used to predict SSC in the testing phase in all the watersheds. Watershed static attributes were used as predictor variables along with the dynamic variables.
(3) Ungauged models: All the 167 USGS stations were randomly divided into 3 groups, with each but one group containing 56 stations. Three global models were developed such that each model was trained on the watersheds contained in the two groups and tested using testing period data from the remaining group. Meyer et al. (2018) referred to this strategy as leave-location-and-time-out strategy. Fig. 2 illustrates this modeling strategy.

Further, each model was developed using the four sets of dynamic variables to comprehensively assess the effects of simulated SY and streamflow on the model results:

(1) Baseline model: The precipitation, the minimum and maximum temperatures, the solar radiation, and the LAI were used as the dynamic variables. This model represents the baseline case where only dynamic meteorological variables and static attributes were used to predict SSC.
(2) Model with simulated SY as a predictor: The daily SY simulated using the MMF model were used along with the baseline dynamic variables.
(3) Model with simulated streamflow (Q) as a predictor: The Q simulated using the VIC runoff and kinematic wave routing were used along with the baseline dynamic variables. A few models were also developed using the observed Q instead of simulated Q to understand the impact of simulated Q errors on the model performance.
(4) Model with both simulated SY and Q: The simulated SY and Q were used along with the baseline dynamic variables.

3.2. LSTM model development

3.2.1. Hyperparameter optimization

Several hyperparameters have to be specified by the user to develop an optimal model, including the number of LSTM layers, hidden dimension (h_{dim}), sequence length (L, number of time-steps in the im-

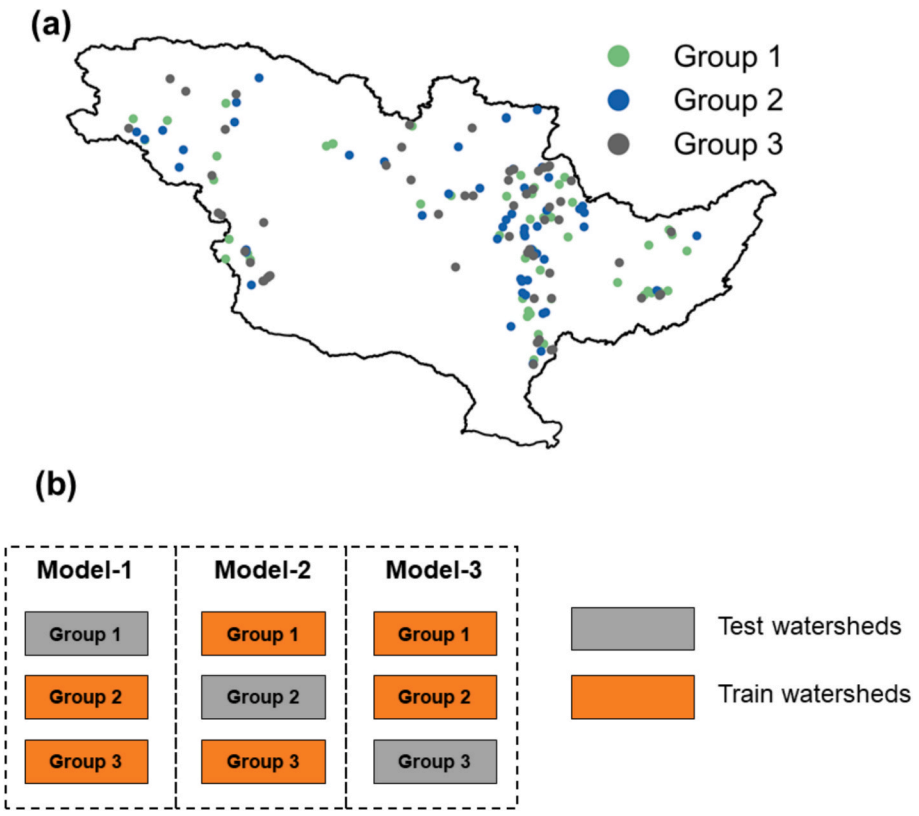

Fig. 2. Experimental setup for ungauged scenario. (a) Locations of the USGS stations in groups 1, 2, and 3. (b) Leave-location-and-time-out modeling for model performance assessment in the hypothetical ungauged scenario.

mediate past to be used as predictors), number of epochs (e), learning rate (α), and the objective/loss function (*Loss*) to be minimized. A single-layer LSTM model was used in this study. The other hyperparameters were determined using the hold-out validation scheme. The training period data from each watershed was divided into two subperiods: subtraining (the first 67 % of the training data) and validation (the remaining 33 % of the training data). Several LSTM models were trained with different combinations of hyperparameters (see Table B1) on the subtraining period data, and NSEs were computed for all the watersheds by comparing observed and simulated SSC during the validation period. Fig. B1 shows the 25th (NSE25), 50th (NSE50), and 75th (NSE75) percentile NSEs for different hyperparameter combinations. Several combinations resulted in comparable NSEs. The learning rate of 0.001 gave the optimal performance in all the cases. The choice of other parameters was unclear. The number of training epochs (e) had different impact in different watersheds: Increasing e from 20 to 40 increased NSE25 (Fig. B1, top panel); however, increasing e beyond 20 decreased NSE75 (Fig. B1, bottom panel). Therefore, $e = 20$ was used as the early stopping criteria to avoid overfitting. The choice of h_{dim} and L was also unclear as several combinations of h_{dim} and L yielded comparable values of NSE. The values $L = 30$ and $h_{\text{dim}} = 128$ were used to develop relatively parsimonious LSTM models. In what follows, the results for the models developed using the following hyperparameters are discussed: $h_{\text{dim}} = 128, L = 30, e = 20, \alpha = 0.001$. The NSE25, NSE50, and NSE75 with this combination were −0.03, 0.23, and 0.42, respectively. The maximum NSE25, NSE50, and NSE75 were −0.002, 0.26, and 0.45, respectively, obtained using different hyperparameter combinations. These parameters were used to train three types of models on the entire training data: local, gauged, and ungauged.

3.2.2. Parameter optimization

The model parameters were trained using the stochastic gradient descent algorithm implemented using the Adam optimizer with a learning rate of 0.001, momentum parameters of 0.9 and 0.999, and a batch size of 32. The model parameters were trained using 8 seeds to account for randomness in the training procedure. The results are presented for the predictions obtained by averaging the predictions of the 8 models.

3.2.3. Standardization of predictor variables

The predictor variables were standardized differently for local and global models. In the local models, each predictor variable was scaled by its maximum value. In the global model, each predictor variable was mean-centered and scaled by the standard deviation. Typically, the second approach is used in ML, but the first approach yielded slightly better results in the case of local modeling. All the standardization statistics were computed only on the training data, which were then used to standardize both the training and testing data. The standardization method of SSC values was also different in local, global, and ungauged models. In the local models, the SSC values in each watershed were divided by the corresponding maximum observed value during the training period. In the global gauged models, the SSC values of each watershed were divided by the corresponding training period standard deviation. In the ungauged models, the SSC values of each watershed were divided by a common standard deviation computed by concatenating the training period SSC data from all the training watersheds. For example, when the data from group 1 were used for testing in the ungauged scenario, the standard deviation of SSC obtained by concatenating the training period SSC values from the watersheds in groups 2 and 3 was used as the standardization constant.

3.2.4. Objective (loss) function

To analyze the effect of the loss function on model performance, the models were trained by minimizing two different objective functions. The first objective function was the variance normalized mean square errors (*NMSE*), which is equivalent to (1 −*NSE*) where *NSE* stands for Nash-Sutcliffe efficiency (Nash and Sutcliffe, 1970):

$$NMSE_t = \sum_w \frac{MSE_w}{\sigma_w^2}, \quad (2)$$

where MSE_w and σ_w denote the unnormalized mean-squared errors and standard deviation of SSC in the watershed w in the training period, respectively. Here, $1/\sigma_w^2$ acts as the normalization constant for watershed w. The second objective function used was normalized mean absolute errors (*NMAE*):

$$NMAE_t = \sum_w \frac{MAE_w}{\sigma_w}, \quad (3)$$

where MAE_w denotes unnormalized mean absolute error of the watershed w. Using NMSE (or NSE) as the objective function yields the conditional weighted mean as the optimal prediction, and using NMAE as the objective function yields the conditional median as the optimal prediction. The conditional median can be a more robust estimate because it is robust to extreme values in the conditional distribution. Note that the explicit normalization of the loss function was not required for the local and global models. In the case of local modes, the models were trained on individual watersheds. In the global models, the SSC values were standardized by corresponding standard deviations, eliminating the need for loss function normalization.

To understand the conditional distribution of SSC estimated by the LSTM model, the quantile loss function was used (Koenker and Bassett, 1978):

$$qloss_t(\tau) = \sum_w \frac{qloss_w(\tau)}{\sigma_w} \quad (4)$$

$$qloss_w(\tau) = \sum_{j=1}^{T} (1-\tau)\left(\hat{y}_j - y_j\right) I\left(\hat{y}_j - y_j\right) + \tau\left(y_j - \hat{y}_j\right) I\left(y_j - \hat{y}_j\right), \quad (4a)$$

where τ denotes the quantile to be predicted, y_j and $\hat{y}_j$ denote the observed and predicted value, I denotes the indicator function, which takes value of 1 if the argument is positive and 0 otherwise, and T denotes the number of observations. In this study, $\tau = 0.10$ and 0.90 were used to identify the lower and upper limits of the distribution.

3.2.5. Determination of the importance of a predictor variable

The importance of different predictor variables was identified by randomly permuting the variables one-at-a-time, and computing the decrease in NSE value after permutation. The decrease in NSE was computed separately for all the watersheds. A larger decrease in NSE after the permutation of a predictor suggests that the predictor variable is of higher importance for SSC simulations.

3.3. Model evaluation

NSE (Nash and Sutcliffe, 1970) and Kling-Gupta Efficiency (KGE; Gupta et al., 2009) were used to evaluate the models developed. NSE describes the amount of observed variability captured by the model:

$$NSE = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2}, \quad (5)$$

where y_t, and $\hat{y}_t$ denote the observed and simulated SSC at timestep t, and $\bar{y}$ denotes the observed mean SSC.

KGE combines three different performance measures to evaluate the model performance:

$$KGE = 1 - \sqrt{(1-r^2) + (1-\alpha)^2 + (1-\beta)^2}, \quad (6)$$

$$\alpha = \frac{\mu_{sim}}{\mu_{obs}}; \beta = \frac{\sigma_{sim}}{\sigma_{obs}}, \quad (6a)$$

where r, α, β denote the correlation coefficient, the bias in mean, and the bias in standard deviation between the simulated and the observed time series, respectively, μ and σ denote the mean and the standard deviation, respectively. To further understand the model performance, the bias in peak and the bias in peak timing errors were also computed. The ideal values for α, and β are 1. The values less than 1 indicate underestimation of mean (α) and variability (β). Similarly, the values greater than 1 indicate overestimation of mean (α) and variability (β).

To further evaluate the LSTM's skill in simulating SSC, the NSE and KGE values obtained using the LSTM were compared against the values obtained from the SRC developed using simulated streamflow. SRC for each station was developed using the equation:

$$SSC = aQ_{sim}^b \quad (7)$$

where a, b, and Q_{sim} denote the model parameters and simulated streamflow, respectively. The model parameters were estimated using the training data, and NSEs and KGEs were computed on the testing data. The parameters of SRC were estimated using the ordinary least square regression between logarithms of Q and SSC. The simulated *SSC* values were converted back to original space by taking exponential of the simulated log*SSC* values and, subsequently, by multiplying it by a correction factor of $\exp\left(\frac{\sigma^2}{2}\right)$. Here, σ^2 denotes the variance of residuals between observed and predicted SSC. The correction factor compensates for systematic underprediction of SSC in the antilog space (Ferguson, 1986). The performance metrics were computed in the antilog space. Since the SRC is developed using simulated Q, it will be denoted by SRCsim. The LSTM model can be considered good only if it performs better than the SRCsim.

4. Results

4.1. Predictability of SSC using NMSE as the objective function

The cumulative distribution functions (CDFs) of NSE values obtained from different models are shown in Fig. 3a. CDF plots provide an overall summary of the performance of different models across all the 167 watersheds. Further, KGE and its three components were also analyzed to assess the ability of the LSTM models in terms of simulating the SSC dynamics (indicated by correlation coefficient and bias-in-standard-deviation), and sediment yield (indicated by bias-in-mean) (Fig. 3b). Table 2 lists the 25th and 75th percentiles, mean, and median of NSEs across different watersheds and the number of watersheds with positive NSE.

The performance of the LSTM models was mixed across different watersheds. The local and global LSTM performed better than the SRCsim in most watersheds. Among the three KGE components, the correlation coefficients were typically high and the biases were poor. The NSE values were positive in about 78 %, 70 %, and 50 % of the watersheds for global, local, and ungauged models, and the NSEs were less than 0.5 in ≈ 90 −95% of the watersheds in the gauged, local, and ungauged scenarios. The correlation coefficients were positive for most of the watersheds and greater than 0.40 for 40–60% of the watersheds (Fig. 3b). The mean SSC was overpredicted in 60% of the watersheds, and the variance in SSC was underpredicted in 70 −80% of the watersheds by the local, global, and ungauged models (Fig. 3b).

The NSEs obtained by the SRCsim were less than 0.20 in 80 % of the watersheds, suggesting that using only the VIC simulated streamflow and power law relationship between SSC and Q resulted in poor simulations of SSC. Therefore, SRCsim is not a strong benchmark, but we discuss it in the paper as the SRC is a popular SSC simulation tool. The local, global, and ungauged models performed better than SRCsim in 70, 80, and 30 % of the watersheds. The LSTM models yielded higher correlation coefficients than the SRCsim in most watersheds (Fig. 3b). The SRCsim tended to underpredict mean SSC compared to the LSTM models (Fig. 3b), which resulted in a better NSE in some watersheds where the LSTM models significantly overpredicted the SSC.

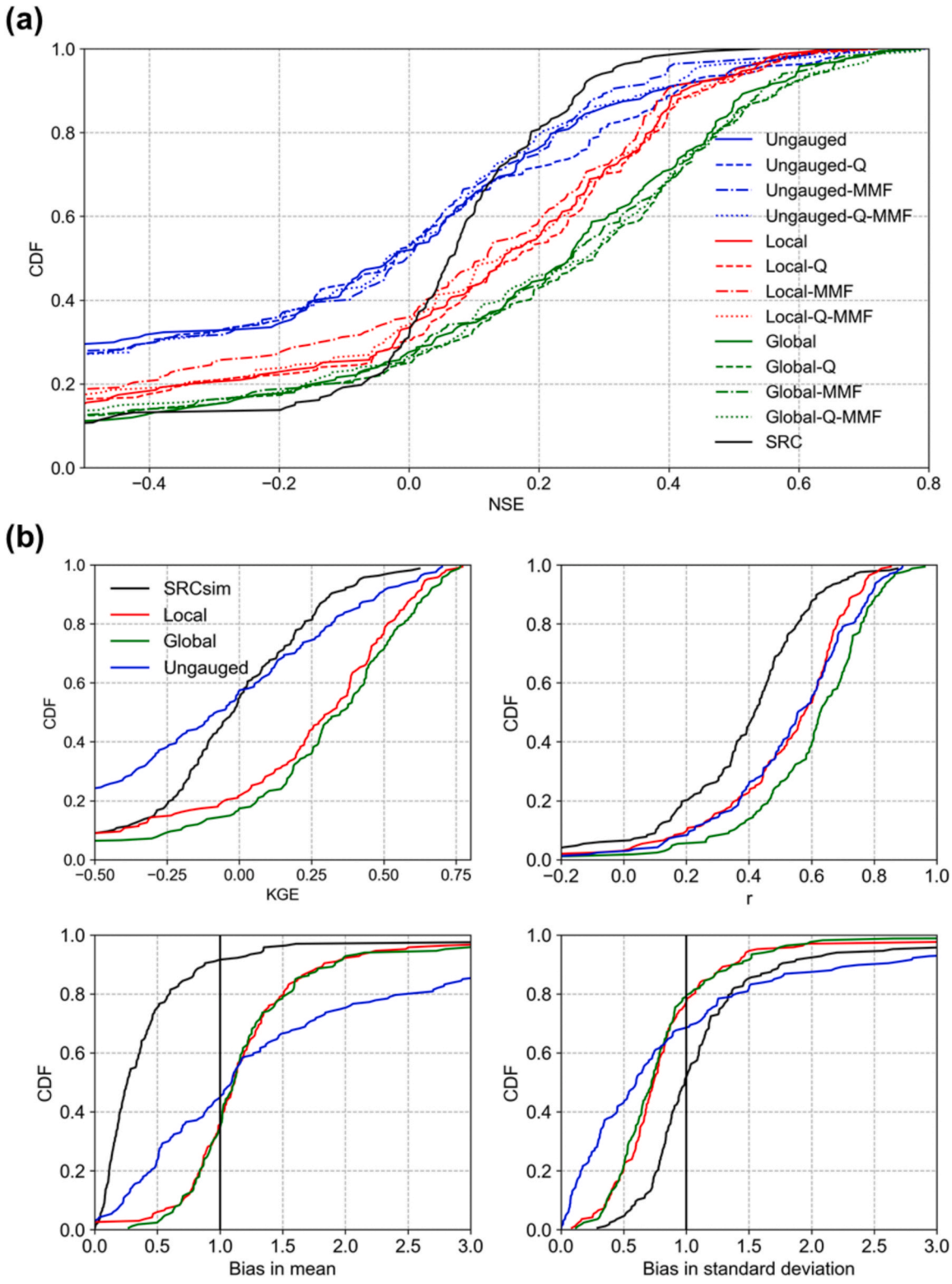

Fig. 3. (a) CDF plots of NSE values obtained using different models, using NMSE as the objective function and DAYMET precipitation data. The symbols Q and MMF indicate that simulated streamflow and MMF simulated sediment yield were used as the dynamic predictor variables along with the baseline dynamic variables. The absence of these symbols indicates that none of these two variables were used as predictors. (b) CDF plots of KGE and its components obtained from different models using NMSE as the objective function, DAYMET precipitation data, and simulated Q as one of the predictors.

The performance of the global models was better than the performance of the local models, implying that data from donor watersheds add useful information for the prediction of SSC (here, the term 'donor watersheds' is used to refer to the watersheds other than the watershed where the model is used to simulate SSC). The performance was the worst in the ungauged scenario, but relatively good performance was achieved for a small percentage of the watersheds (e.g., NSE $>$ 0.26 and 0.40 for 25 % and 10 % of the watersheds, respectively). The ungauged

Table 2
25th percentile, mean, median, 75th percentile of NSE values and the number (percentage) of watersheds with NSE > 0 for different models.

	Local models				Global models				Ungauged models			
	B	**B-Q**	**B-MMF**	**B-Q-MMF**	**B**	**B-Q**	**B-MMF**	**B-Q-MMF**	**B**	**B-Q**	**B-MMF**	**B-Q-MMF**
Objective function: Normalized Mean Square Error (NMSE)												
25th	−0.061	−0.057	−0.277	−0.117	−0.014	0.008	−0.007	−0.004	−0.958	−0.558	−0.715	−0.605
mean	−2.156	−3.639	−4.542	−3.689	−1.184	−1.231	−1.251	−2.547	−193.128	−183.484	−161.36	−221.115
median	0.147	0.168	0.121	0.144	0.249	0.286	0.253	0.277	−0.022	−0.019	0.001	−0.03
75th	0.345	0.351	0.33	0.34	0.427	0.44	0.437	0.434	0.186	0.261	0.199	0.174
Watersheds with NSE > 0	113	118	109	112	122	127	124	125	82	80	84	79
	(67 %)	(70 %)	(65 %)	(67 %)	(73 %)	(76 %)	(74 %)	(75 %)	(49 %)	(48 %)	(50 %)	(47 %)
Objective function: Normalized Mean Absolute Error (NMAE)												
25th	0.032	0.044	0.001	0.013	0.064	0.079	0.068	0.079	−0.702	−0.673	−0.593	−0.522
mean	−1.369	−2.342	−0.838	−3.743	−0.314	−1.025	−0.219	−0.763	−162.632	−122.396	−91.862	−107.271
median	0.167	0.192	0.158	0.196	0.276	0.297	0.275	0.295	−0.023	−0.018	−0.005	0.002
75th	0.362	0.384	0.345	0.347	0.464	0.473	0.459	0.454	0.189	0.262	0.246	0.277
Watersheds with NSE > 0	137	134	127	128	137	139	140	136	79	79	81	84
	(82 %)	(80 %)	(76 %)	(76 %)	(82 %)	(83 %)	(84 %)	(81 %)	(47 %)	(47 %)	(48 %)	(50 %)

B: Models with baseline predictor variables.
B-Q: Models with baseline predictor variables and simulated Q as predictor variable.
B-MMF: Models with baseline predictor variables and MMF simulated SY as predictor variable.
B-Q-MMF: Models with baseline predictor variables, simulated Q and MMF simulated SY as predictor variables.

model had larger biases in the simulated mean and standard deviation. This suggests that the LSTM model trained on donor watersheds could simulate the temporal variation in SSC (indicated by the correlation coefficient, *r*), but could not accurately simulate the sediment yield (indicated by biases in mean). Notably, the local and global models had similar performance in terms of bias-in-mean and bias-in-standard-deviation. However, the local model had smaller *r* values compared to the global model, indicating that the global model could simulate the SSC temporal variation better than the local model.

In all three modeling scenarios (i.e., global, local, and ungauged), the models developed using the simulated Q as one of the predictor variables were typically the best (Fig. 2), even though simulated Q only slightly improved the model performance in the majority of the watersheds (Table 2). For example, the median NSE increased from 0.147 to 0.168 in local modeling experiments, from 0.249 to 0.286 in global modeling experiments, and from −0.22 to −0.19 in global ungauged modeling experiments (Table 2) by adding simulated Q as a predictor variable. The mean NSE values, however, decreased in the local and

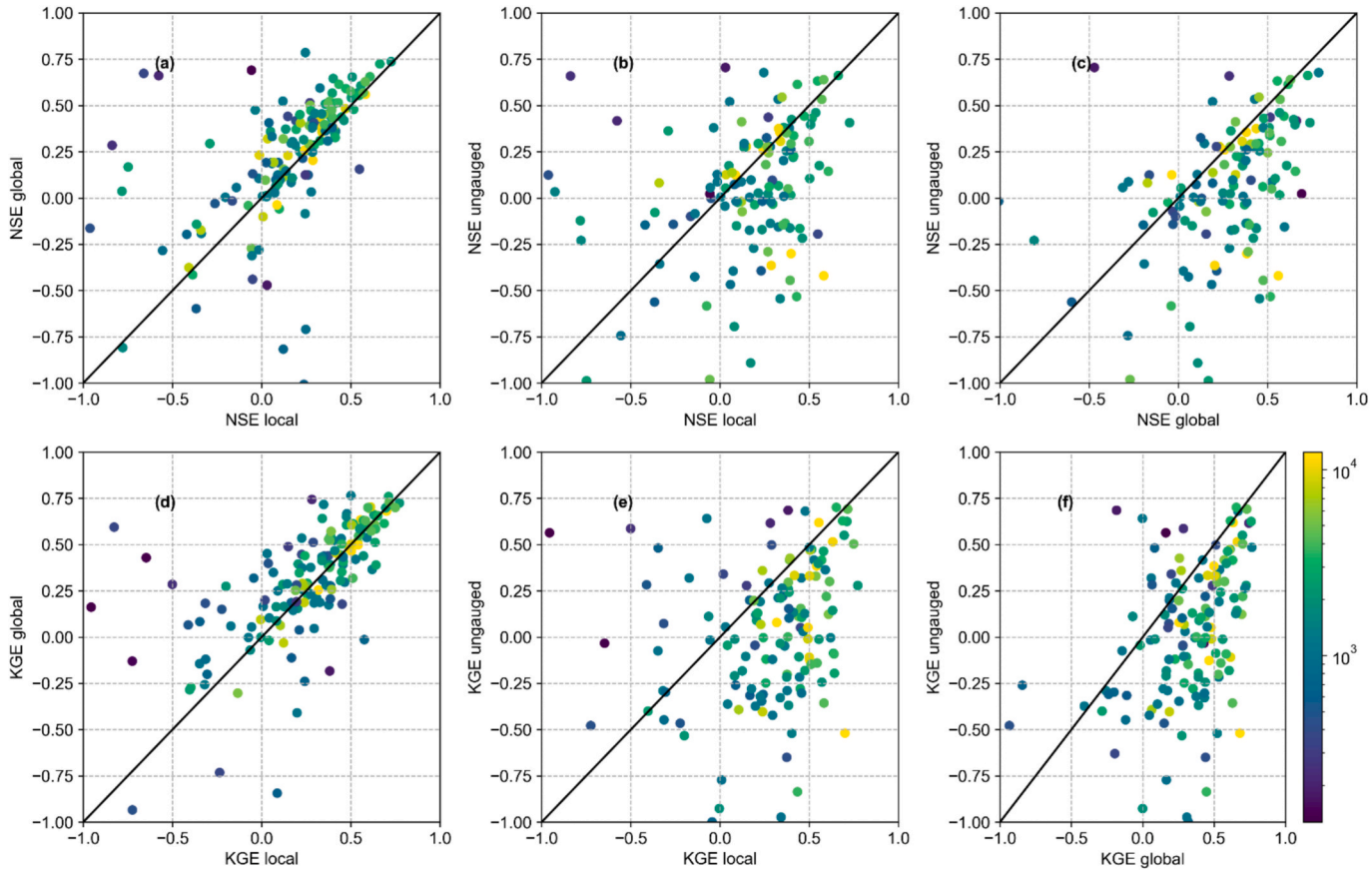

Fig. 4. Comparison of NSEs (top panel) and KGEs (bottom panels) obtained using different models. The colors in the plots represent the number of daily SSC observations in the watersheds. These models were developed by using Q as one of the predictor variables.

global scenarios by using simulated Q as an additional predictor variable, suggesting that the simulated Q worsened the performance in the low NSE region. Using the sediment yield simulated by the MMF model as one of the predictors did not help improve the model performance in any of the three modeling scenarios (Fig. 2, Table 2).

While the CDF plots provide a good overall summary of the model performance, they do not provide much information about the comparative performance of models in individual watersheds. Fig. 4 compares the NSEs and KGEs obtained by different models for individual watersheds. No single model was the best for all the watersheds. The local models yielded higher NSE and KGE compared to the global model for 30% and 39% of the watersheds, respectively (Fig. 4a, d). The ungauged models yielded higher NSEs and KGEs compared to the local model in 34% and 22% of the watersheds, respectively (Fig. 4b, e). The ungauged models yielded higher NSEs and KGEs compared to the global model in 21% and 14% of the watersheds, respectively (Fig. 4c and 4 g). One can conjecture that the poor performance of the local models compared to the ungauged models in a few watersheds was due to a limited amount of training data available to the local models. However, the watersheds where the ungauged model performed better had various lengths of training data (from 100 to 1000 days), implying that the better performance in ungauged scenarios in these watersheds was not just because of the smaller length of training data available to the local models.

The spatial distributions of NSEs and correlation coefficients for local, global, and ungauged models were plotted to get insights into the relationship between model performance and watershed attributes (Fig. 5). While the model performances were mixed across the MRB, some general patterns could be observed. The NSEs were typically poorer in the watersheds located in the arid MB and ARW regions (NSE

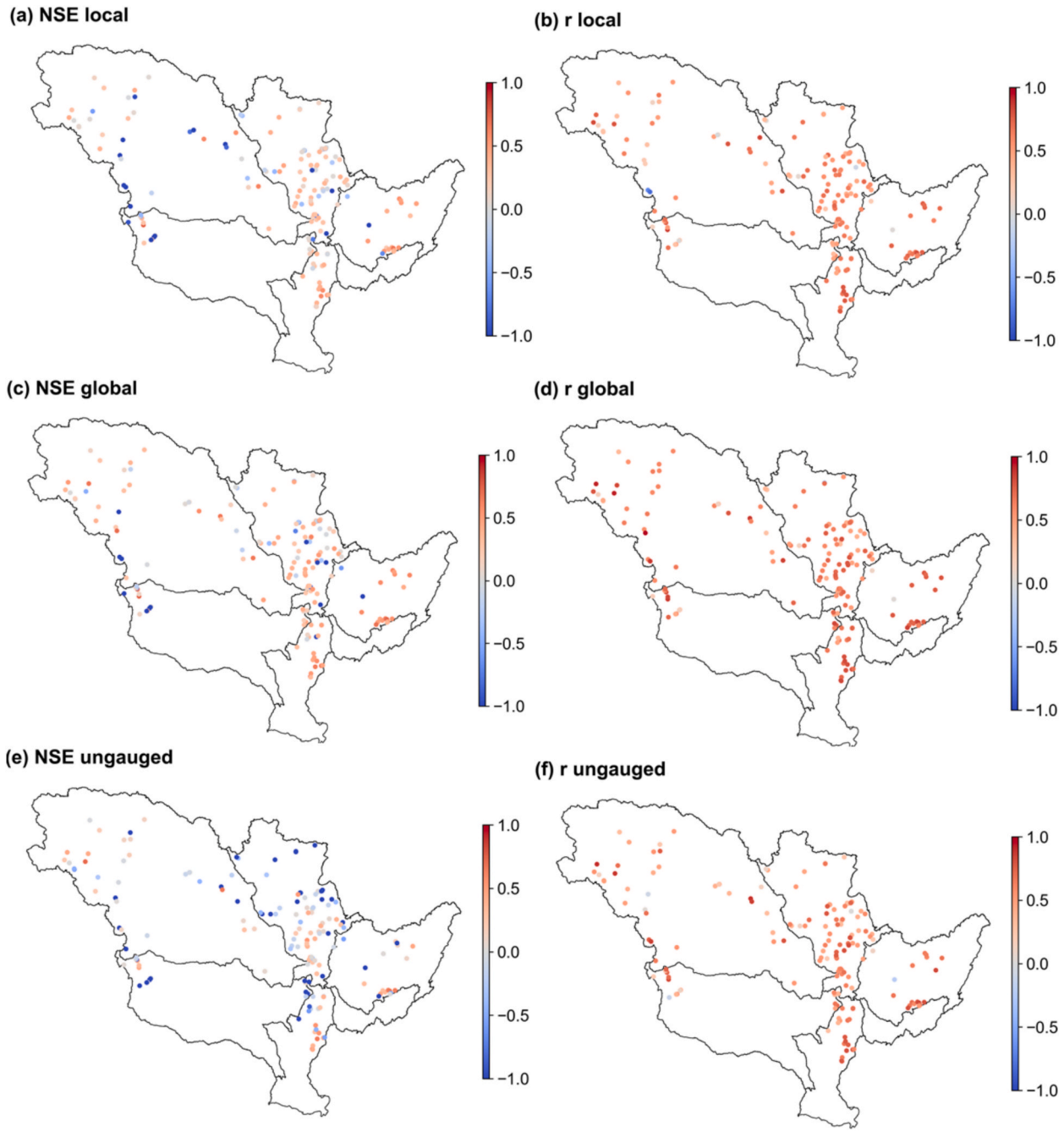

Fig. 5. Spatial distributions of NSE (left) and correlation coefficient (r; right) values for the models trained in the three modeling scenarios.

> 0 for 64.6 % of the watersheds in these regions with the median NSE = 0.17 using the global model). The local models performed significantly worse than the global models in these two regions. In the moderately wet UMRB, the NSEs were mixed (NSE > 0 for 70 % of the watersheds with the median NSE = 0.22 using the global model). The local and global models performed similarly in this region. In the ORB and LMRB, the wettest regions in MRB, the highest NSEs were obtained (NSE > 0 for 89.7 % of the watersheds with the median NSE = 0.48 using the global model). The local models performed worse than the global model in these regions. The ungauged models performed worse than the global and local models in all the regions.

The results discussed above clearly show that the SSC simulations were more accurate in the wetter regions of the MRB (see also Fig. E2 in Appendix E). Among static watershed attributes, only annual precipitation and forest fraction were clearly related to the model performance (Fig. E2, Appendix E): Increasing annual precipitation and forest fraction increased the model performance. Since forest fraction is correlated with annual average precipitation, these results suggest a strong control of the areal average precipitation on riverine SSC in the wetter regions. In the dryer regions (MB and ARW), the spatial variability of precipitation and other watershed attribute also significantly affected the SSC, resulting in a strongly non-linear relationship between predictor and

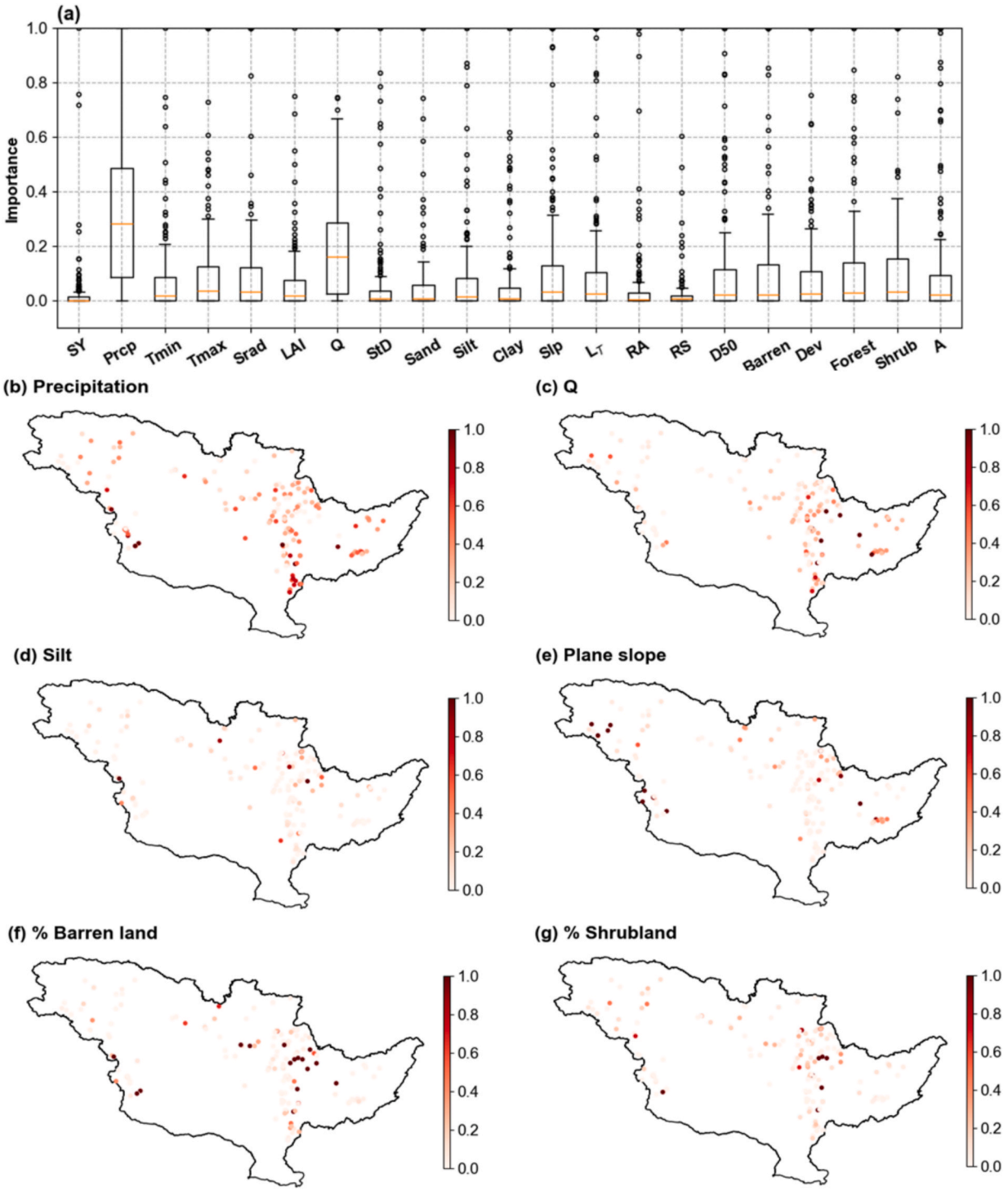

Fig. 6. Importance of different predictor variables in simulating the SSC by the global LSTM model.

response variables. The weak correlations between observed Q and SSC in the MB region (see Fig. C1, Appendix C) also indicate strongly nonlinear SSC dynamics in this region. Fig. E2 (Appendix E) also shows that performance was often better in the watersheds with higher correlations between the *observed* Q and SSC. Further, number of SSC measurements in a watershed also affected the model performance because the extremely poor performance metrics (e.g., large biases) were only observed in the watersheds with a small number of SSC measurements (Fig. E3; Appendix E).

Fig. 6 shows the importance of each predictor variable as identified by the LSTM model. Note that a variable's importance indicates its role in developing the LSTM model and depends strongly on the data quality. If a variable is unimportant according to this analysis, it does not necessarily mean that the variable does not control the SSC dynamics in the real world. Among the dynamic variables, the precipitation was the most important predictor variable, followed by the simulated Q (Fig. 6a). Other dynamic variables, such as solar radiation, T_{min}, and T_{max}, were slightly important. The precipitation was more important in the wetter regions of the MRB (ORB, URRB, LMRB; Fig. 6b) than in the dryer regions, again indicating that precipitation has stronger control over the SSC dynamics in the wetter regions. Among the static variables, the land use type (fractions of barren and shrubland), the plane slope (denoted by 'Slp' in Fig. 6a), and the soil texture were important. The plane slope was very important in the southern ORB region and some of the high-sloped watersheds in the western MRB (Fig. 6e). The variables, such as the reservoir parameters and the watershed area, were not important.

The significantly lower importance of the simulated Q compared to the observed precipitation might be surprising at first; it is, however, explained by the fact that the simulated Q contains large errors compared to the observed Q in many watersheds (Fig. C1, Appendix C). Still, the simulated Q was an important predictor in the ORB and LMRB regions, where the simulated Q was more accurate (Fig. C1). Similarly, the MMF-simulated SY was unimportant because it was a poor estimator of true SY (e.g., Tan et al., 2018; Li et al., 2022). The low importance of dam-related variables might be due to the inadequacy of these variables in terms of representing the effects of dams. Alternatively, it might be because most of the dams in the MRB were built before 1983, and the LSTM might have identified the effects of dams as the behavior of SSC dynamics in the dammed watersheds.

4.2. Impact of objective functions on the performance of the trained models

Different objective functions emphasize different parts of the modeled time series, and the DL model extracts different kinds of information from available data depending on the objective function used. Therefore, the LSTM models were trained using NMSE and NMAE as objective functions to test their impacts on the model performance. Fig. 7 and Table 2 compare different models trained using NMSE and NMAE as the objective functions. Overall, the higher NSEs were obtained using NMAE as the objective function, especially for the watersheds where SSC was poorly predicted. Similar results, however, were not seen for the KGE values. In the ungauged scenario, the NMAE- and NMSE-trained models yielded similar KGEs. In the global scenario, the KGEs were only slightly better for NMAE-trained models. In the local scenario, NMAE-trained models yielded lower KGEs for most watersheds because of larger biases in SSC simulations (results not shown). These results show that the best objective function to train the LSTM model depends upon the watershed considered, the training procedure (local,

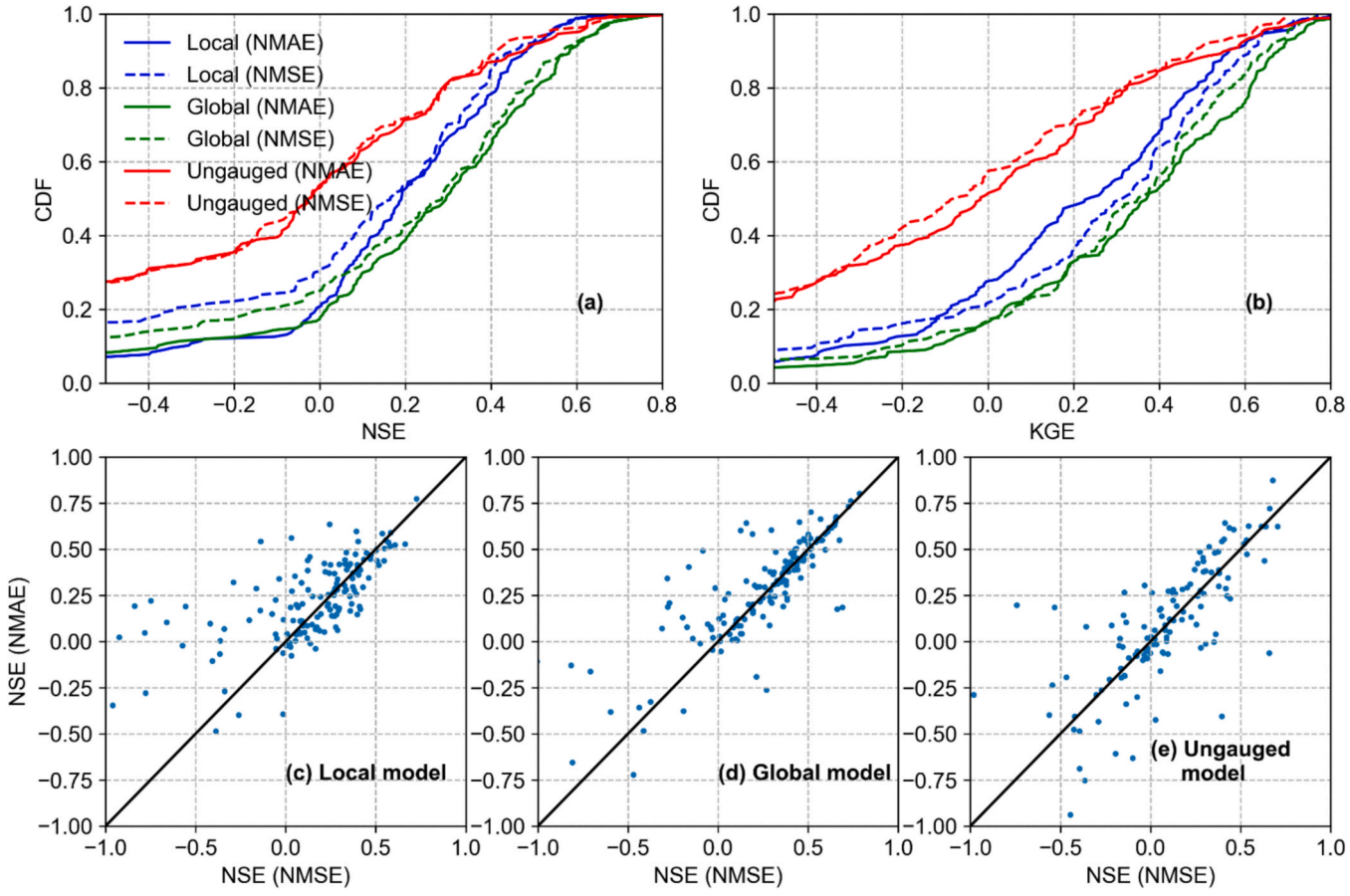

Fig. 7. Comparison of NSE and KGE values obtained using NMSE and NMAE as the training objectives in different models using DAYMET precipitation data and simulated Q.

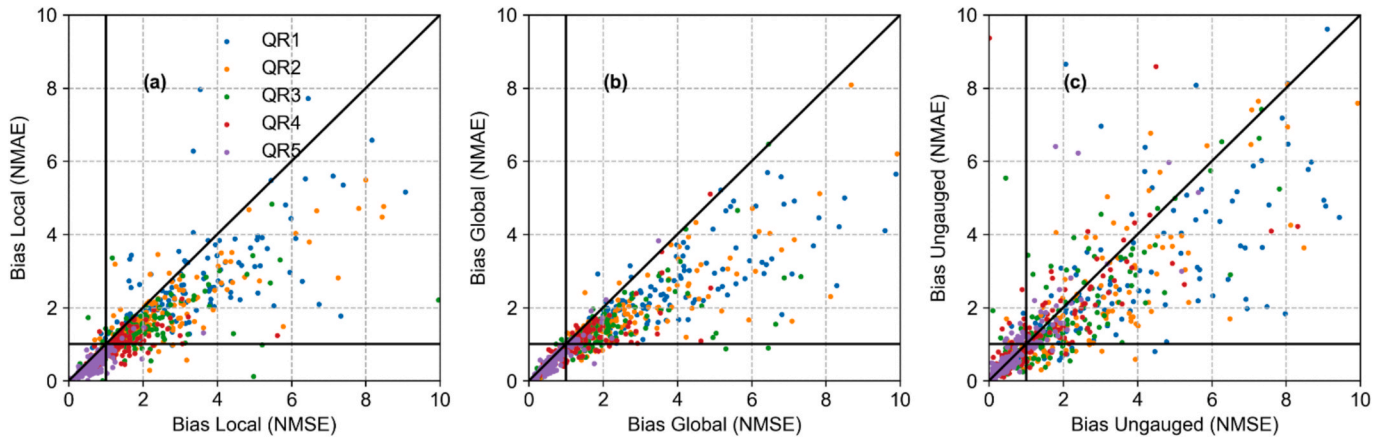

Fig. 8. Comparison of biases-in-mean for five different percentile ranges of the observed SSC. QR1-QR5 denote the five percentile ranges of observed SSC (see the text related to this Figure). The horizontal and vertical lines denote the ideal values of bias (=1); a good model should fall close to horizontal and vertical lines for NMAE and NMSE models, respectively. A point in left-bottom quadrant means that the biases are smaller than 1 for both the models implying underestimation by both the models; a watershed falling below 1:1 line in this quadrant means that NMAE-model resulted in greater underestimation compared to the NMSE-model, and vice-versa. A point in top-right quadrant indicates that both the model overestimated the SSC concentration; a point below 1:1 in this quadrant means that NMAE model resulted in smaller overestimation compared to the NMSE-model.

global, or ungauged), and the metric used to measure the model performance.

To understand what parts of the SSC time series are better simulated by the NMSE-and NMAE-trained models, the observed SSC time series at a watershed was divided into five percentile ranges: 0–15 percentile (QR1), 15–40 percentile (QR2), 40–65 percentile (QR3), and 65–90 percentile (QR4), and 90–100 percentile (QR5). Fig. 8 shows the biases in the simulated mean (computed using Eq. (6a) for the different ranges. The low SSCs were overpredicted for most watersheds, and the high SSCs were underpredicted. The NMAE-trained models had smaller biases for low concentrations, i.e., smaller overprediction for low concentrations. Similarly, the NMSE-trained models had smaller biases for high

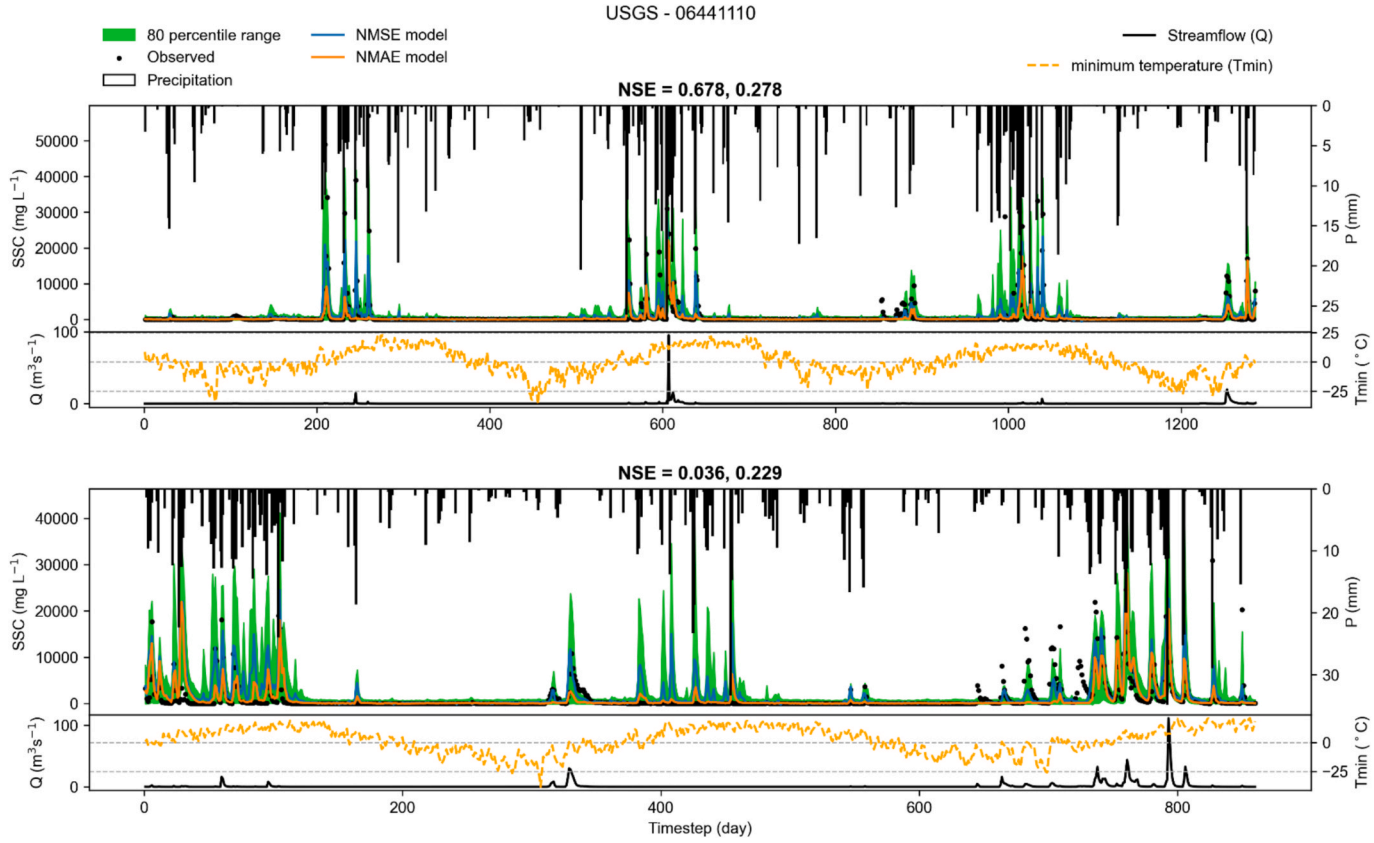

Fig. 9. Observed and simulated SSC by the global models for the training phase (top) and the testing phase (bottom). The blue line, orange line and scatter are SSC simulated by the NMSE model, SSC simulated by the NMAE model and observed SSC, respectively. The green area is the 10–90 percentiles of SSC simulated by the LSTM model using the quantile loss function (Eq. (4). The symbols P, Q, and Tmin denote the observed areal average precipitation (DAYMET), observed streamflow and observed minimum temperature. The NSE values at the top of each subplot are NSEs obtained by the NMSE and NMAE models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

concentrations, i.e., smaller underpredictions of high concentrations. These results suggest that the NMAE- and NMSE-trained models are better suited for low and high SSC simulations, respectively.

4.3. Observed and predicted time series

Figs. 9-12 show the observed and simulated SSC for training and testing phases at four stations to assess the model performance qualitatively. These stations were selected to show different kinds of model behaviors. For the station USGS-06441110 (Fig. 9), training NSE was larger for the NMSE-trained model compared to the NMAE-trained model, but the testing NSE was larger for the latter model. Around the time-step 400 in the testing phase, the observed SSC concentration did not respond to precipitation and was poorly simulated by the NMSE-trained model. The hydro-meteorological conditions for this event were similar to those of other events, but SSC responded to increased precipitation for other such events, suggesting that this event had 'anomalous' SSC dynamics not observed in the training period. One possibility could be the errors in the VIC simulated streamflow, which is partially confirmed by comparing the SSC simulated by the global model trained using simulated and observed streamflow. Fig. F2 (in Appendix F) shows that training the LSTM model on observed streamflow improved the SSC simulations for this event, but the observed SSC peaks were still overestimated.

For the station USGS-06268500 (Fig. 10), the NMSE-trained model yielded larger NSE than the NMAE-trained model in both the training and testing phases. Four of the SSC peaks were completely missed in the testing phase by the LSTM models – these peaks were not even enveloped by the 80-percentile range (the green region), suggesting that the dynamics governing these peaks were not represented in the training phase. The SSC peaks at timesteps 50, 100, and 460 were caused by a small amount of precipitation with a small streamflow response, suggesting that these events may have resulted from short and intense storms with large sediment erosion and transport capacity given that the dominant soil types in this watershed are silt and clay. In addition, spatial locations of the storm's center might have played a role in generating large SSC for these storms, given the large area of this watershed (~1354km^2). Alternatively, these peaks might have occurred due to sediment availability near the watershed outlet.

For the station USGS-05427910 (Fig. 11), a missed peak in the testing phase around timestep 85 was observed. There was a 4–5 day lag between precipitation and the observed SSC peak, and streamflow had a muted response to this event. Such a lag between precipitation and watershed response was not observed for other events, suggesting that observed precipitation was underestimated. The potential underestimation of observed precipitation can also be seen in the training phase around timestep 250.

For the station USGS-03294500 (Fig. 12), the NMSE- and NMAE-trained models overestimated the SSC during the testing phase. This station's training and testing periods had very different hydrometeorological conditions: the training period was cold with high streamflow, while the testing period was warm with low streamflow. This suggests that the bad performance of the LSTM model at this station might be due to the lack of representative training data. Also, there was a dam upstream of this station (built before the study period). Therefore, changes in dam operations might have resulted in changes in riverine SSC dynamics, but the data on dam operations was not available to confirm this hypothesis.

5. Discussion

The results of this study are encouraging because the SSC could be simulated accurately in many watersheds using the LSTM model. However, the LSTM performed significantly worse in the ungauged scenario than in the gauged scenarios. The main difference between the

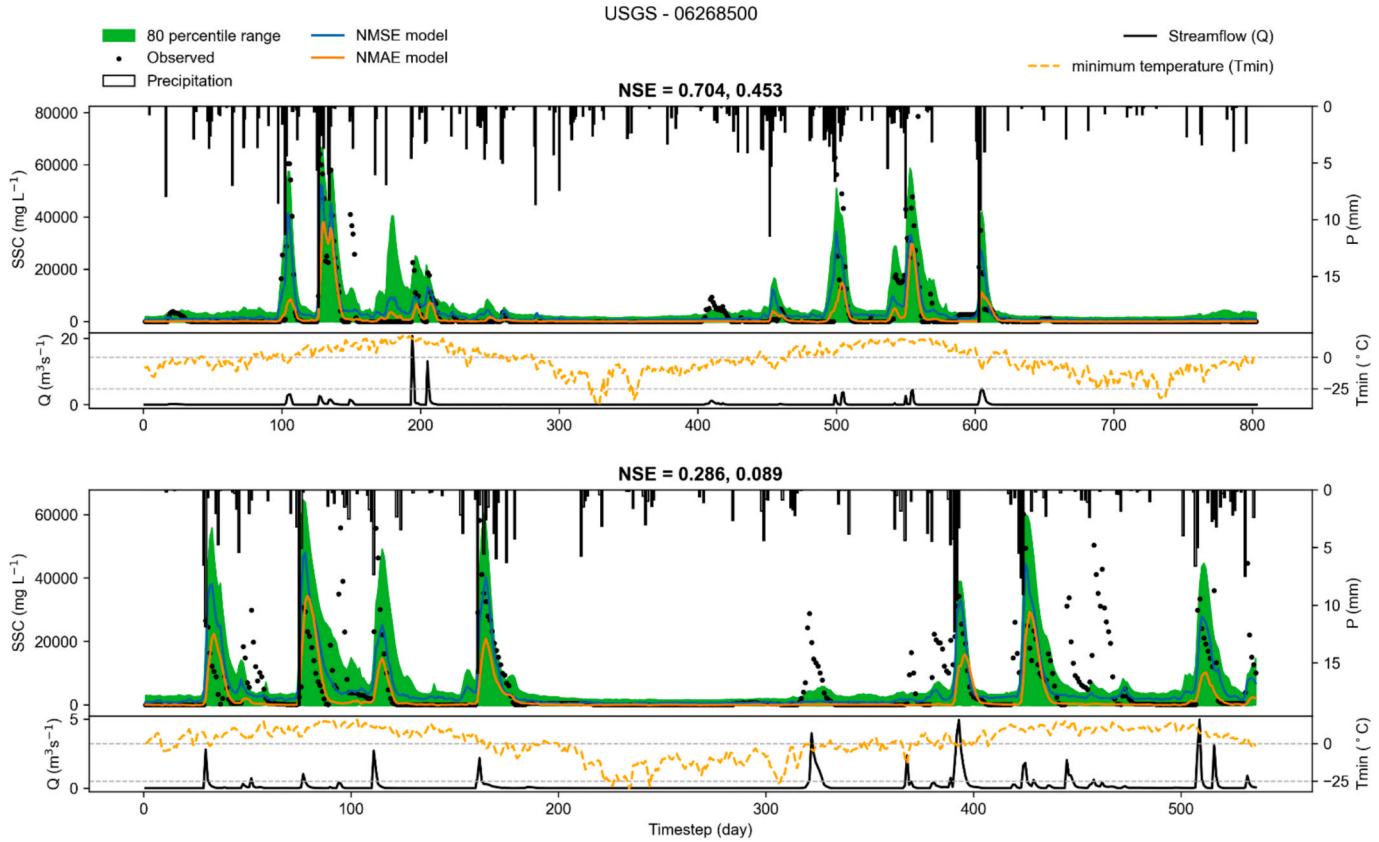

Fig. 10. Same as Fig. 9 but for a different station (USGS-06268500).

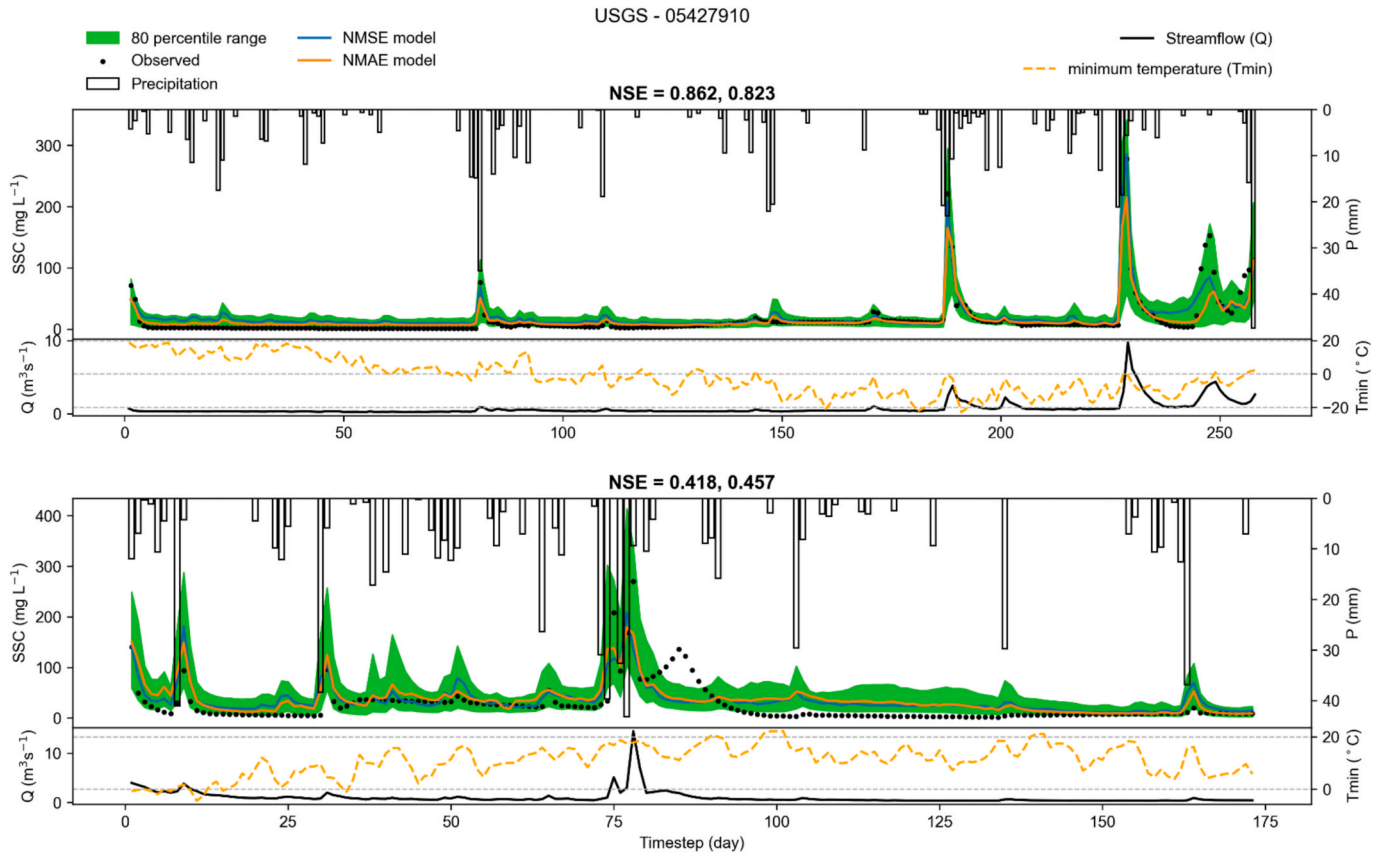

Fig. 11. Same as Fig. 9 but for a different station (USGS-05427910).

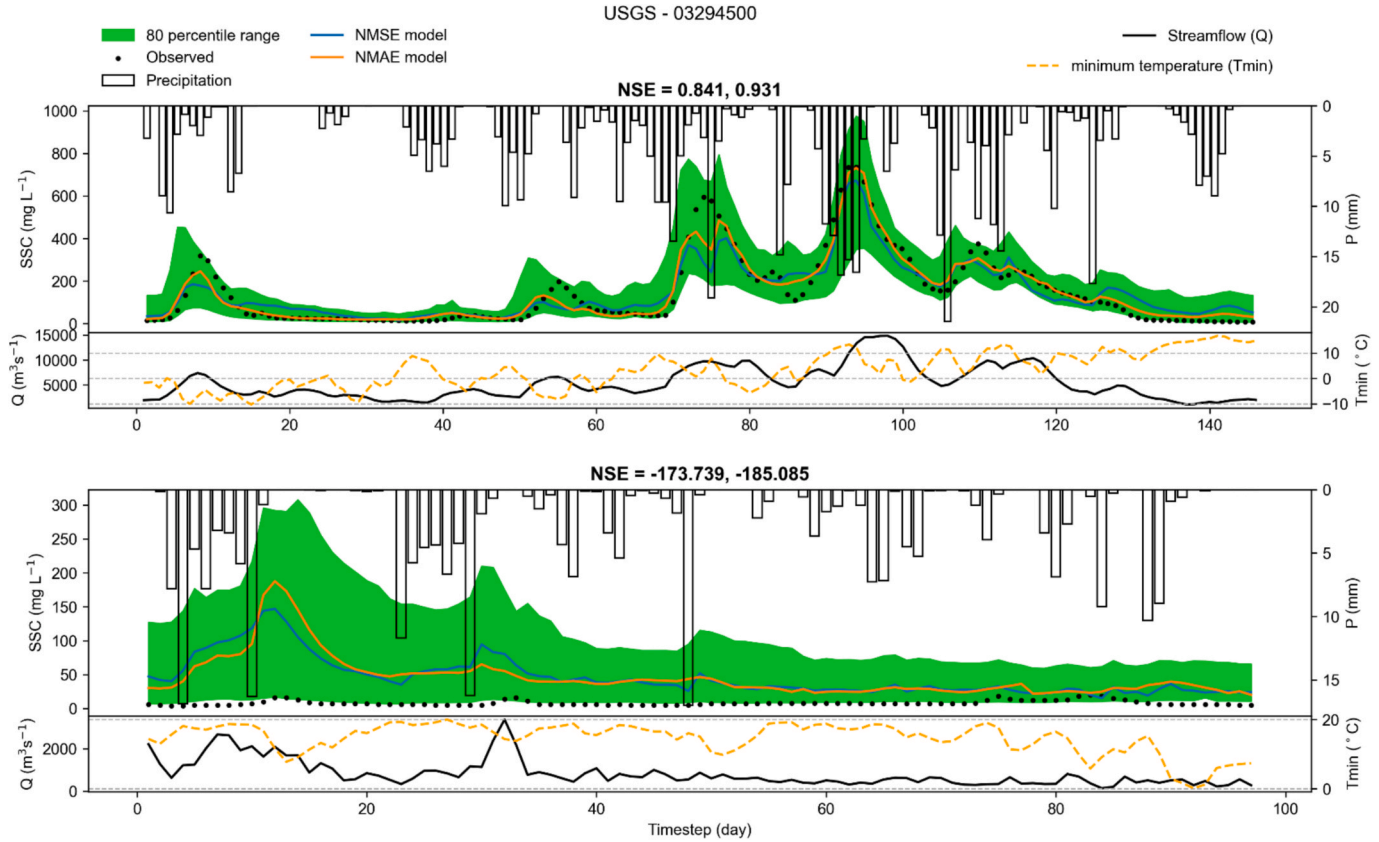

Fig. 12. Same as Fig. 9 but for a different station (USGS-03294500).

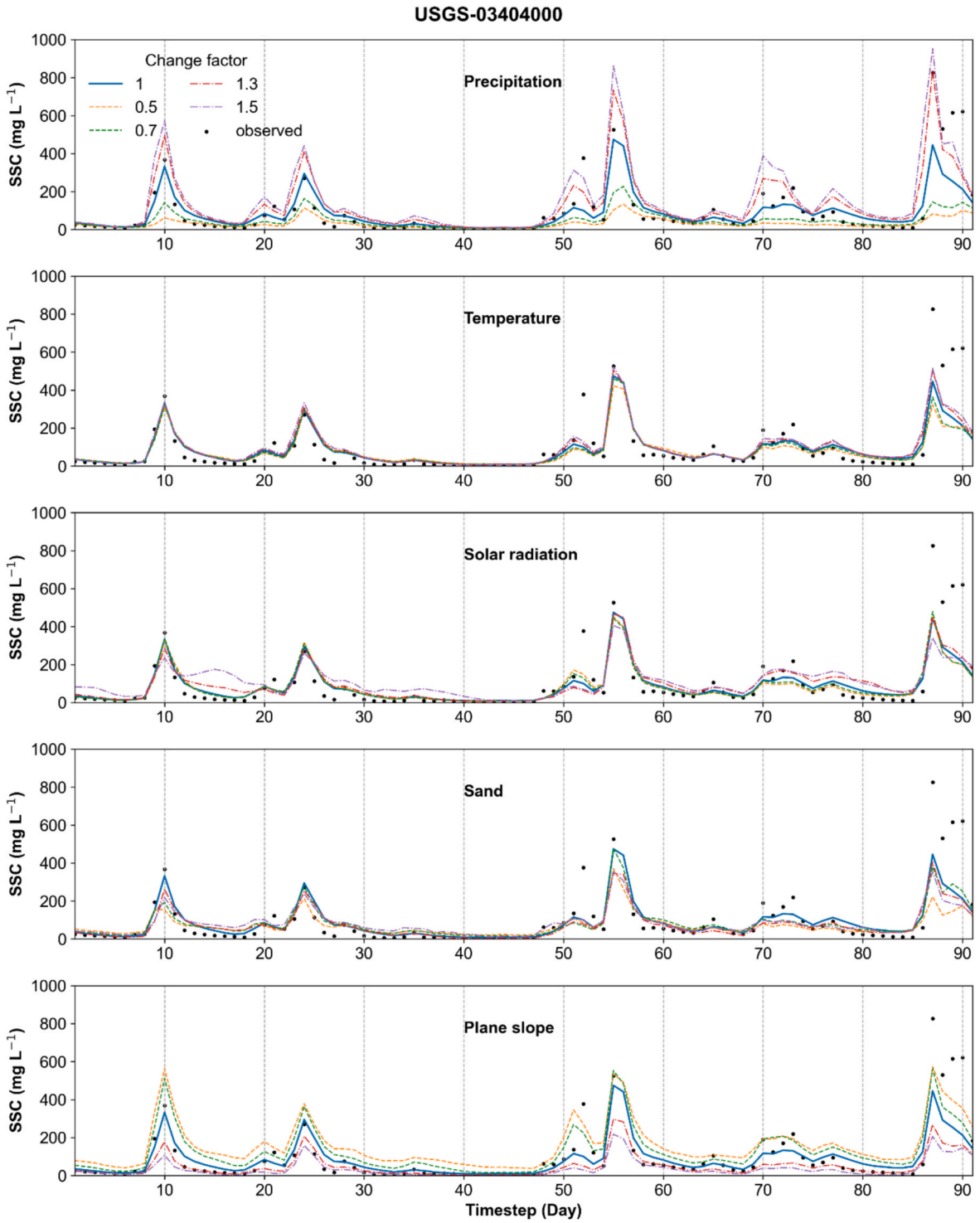

Fig. 13. Relationships learned by the LSTM model. SSC was simulated by the globally trained LSTM model by changing different input variables by different factors.

ungauged and the other two scenarios was that the former had larger biases in the simulated mean. This suggests that the information from donor watersheds help simulate the temporal variation of SSC, but the information from the local watershed is required to better predict the sediment yield. The analysis of the observed and predicted time series (Figs. 9-12 and Fig. E3) shows that one of the reasons for the poor performance of the LSTM models is the lack of data to train the models on all the behaviors of SSC dynamics. For some watersheds, the training period could not include all the hydrometeorological conditions observed in a year. These results illustrate the importance of collecting riverine SSC data. In this context, the trend of a decreasing number of SSC observations in the MRB (Fig. 1) is worrisome.

While the results of this study are not directly comparable with other studies, it appears that the LSTM simulates SSC better than the process-based models. For example, Li et al. (2022) reported negative KGE for monthly SSC for 50 % of their watersheds in the MRB. We expect that the performance of their model will be much worse at the daily scale. Cohen et al. (2014) found large differences in observed and simulated daily sediment flux for five large watersheds in the MRB, but no performance metrics were reported; however, large biases are visible in the Cohen et al. paper.

The SSC was poorly simulated in most of the arid watersheds of the western MRB. Similarly, the streamflow is also difficult to simulate in arid watersheds (e.g., Pilgrim et al., 1988; Coles et al., 1997; McIntyre et al., 2007; Al-Qurashi et al., 2008; Yatheendradas et al., 2008; McIntyre, and Al-Qurashi, 2009; Knoben et al., 2020). One plausible reason is a strong non-linear relationship between the areal average precipitation and SSC in these watersheds, which might, in turn, be due to the spatial variability of precipitation (Lane and Kidwell, 2001) and high temporal variability masked by the aggregation of precipitation at daily scales (Pilgrim et al., 1988). To test if accounting for temporal variability in precipitation improves the prediction accuracy, we also developed the LSTM model by using the 3-hourly MSWEP data. However, no improvements in prediction accuracy were observed. The effects of precipitation spatial variability will also be explored in a future study.

The SSC simulations reported in this study can be improved in three ways. First, using the observed instead of the simulated streamflow improves the SSC simulations in many watersheds (Figs. F1 and F2). Fig. F1 shows that the performance significantly improved in the local and gauged scenarios. In the ungauged scenario, however, only a slight improvement in the performance was obtained by using observed streamflow. Second, the model developed in this study used a single combination of optimal hyperparameters for all the watersheds. However, Fig. B1 suggests that the optimal hyperparameter depends upon the watershed. For example, the optimal number of epochs were 20 for some watersheds, while it was 40 for other watersheds. These results suggest that the global LSTM can be tailored to a specific watershed for better performance in that watershed by hyperparameter tuning, as was conjectured by Gupta and McKenna (2025) for low flow simulations. As an example, Fig. B2 shows the performance of different global models trained using different hyperparameters. Increasing the epochs and the sequence length increased the performance in some low-performing watersheds, but decreased the performance in other watersheds.

Third, the performance of the global models increased when the LSTM was trained on NMAE instead of NMSE objective function. The NMAE-trained models simulated the low concentrations better than the NMSE-trained models, and the NMSE-trained models simulated the high concentrations better. Similar results were reported for streamflow simulations using the LSTM by Gupta and McKenna (2025). These results suggest that if an event can be classified as a low-concentration or high-concentration event (using the information in predictor variables such as precipitation), then NMAE and NMSE predictions should be used for low- and high-concentrations, respectively. This may be explored in a future study.

Among the predictors used, the precipitation, the temperature, the simulated Q, the plane slope, the land-use type, and the soil texture were important variables. The precipitation was more important in the wetter regions than in the drier regions. Further, the performance of the LSTM model was better in the wetter regions. These results suggest that the areal average precipitation had stronger control over SSC dynamics in the wetter regions.

To better understand the relationships identified by the LSTM model, the trained global LSTM model was used to simulate SSC by deterministically changing the values of predictor variables. Fig. 13 shows the effect of changing predictor variable values at one of the USGS stations (similar results were observed for other watersheds). Increasing (decreasing) the precipitation increased (decreased) the SSC. Increasing (decreasing) the maximum and minimum temperatures resulted in a slight increase (decrease) in SSC. The relationship between solar radiation and the SSC was complex. For low-concentration timesteps, increasing the solar radiation increased the SSC. For high-concentration timesteps, increasing the solar radiation increased the SSC for some timesteps and decreased the SSC for other timesteps. Decreasing the plane slope increased the SSC, and increasing the plane slope decreased the SSC; these results contradict the empirical models of sediment yield, where an increase in plane slope is associated with an increase in sediment yield. These results provide some insights into the relationships learned by the LSTM and will be explored in detail in a future study.

6. Conclusions

The following conclusions were drawn from this study:

(1) Global LSTM models, trained on multiple watersheds yielded state-of-the-art performance compared to other modeling approaches that do not use observed streamflow as a predictor. Global models were typically better than local models, underlining the importance of multi-basin training of LSTM.
(2) While the SSC simulations were overall difficult using the LSTM model, the performance was good in some watersheds, with NSE > 0 in 50% and NSE > 0.50 in 10% of the watersheds.
(3) Simulated streamflow (by the VIC model) was only slightly useful for predicting SSC. The daily SY simulated by the MMF model did not add any information to the LSTM model.
(4) Poor performance of the global LSTM model in some watersheds was due to the lack of representation of testing period data in the training period.
(5) The SSC dynamics were strongly controlled by the areal average precipitation in the wetter regions than in the drier regions of the MRB.
(6) The NMAE-trained models were better than NMSE- trained models at simulating low concentrations, and the latter models were better at simulating high concentrations.

7. Research data

All the data generated in this study, including the raw data, model results, and python scripts to run the LSTM models, are available at Gupta and Feng (2025). The Python scripts can also be downloaded from GitHub using the link: https://github.com/AbhinavGupta1611/sed-est.

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CRediT authorship contribution statement

Abhinav Gupta: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Dongmei Feng:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Abhinav Gupta, Dongmei Feng reports financial support was provided by NASA. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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manuscript.

Appendix A. Spatial distribution of dynamic and static attributes

Figs. A1-A3 show the spatial distribution of watershed climatic, soil, land-use, and geomorphological properties. The watersheds in western MRB (the MB and ARW regions) were cold and dry with average annual precipitation of less than 1000 mm and below zero temperatures in the winter months (Fig. A1). The soil texture in these watersheds was dominated by silt and clay and land-use is dominated by shrubs (Fig. A2). The slopes were steep with often large drainage areas (Fig. A3).

The watershed in the ORB and LMRB regions were hot and wet with average annual precipitation of greater than 1200 mm (Fig. A1). Soil texture was dominated by clay and land-use was dominated by forests with some watershed having more than 5 % developed land. Most of the watersheds in the ORB were steep with relatively small drainage areas and smaller channel lengths. The watersheds in the LMRB were flatter with relatively small drainage areas. In the UMRB region, average annual precipitation was between ≈ 1000 −1200 mm (Fig. A1). Soil texture was dominated by sand and the land-use was mixed (Fig. A2). The slopes were mostly flat (Fig. A3).

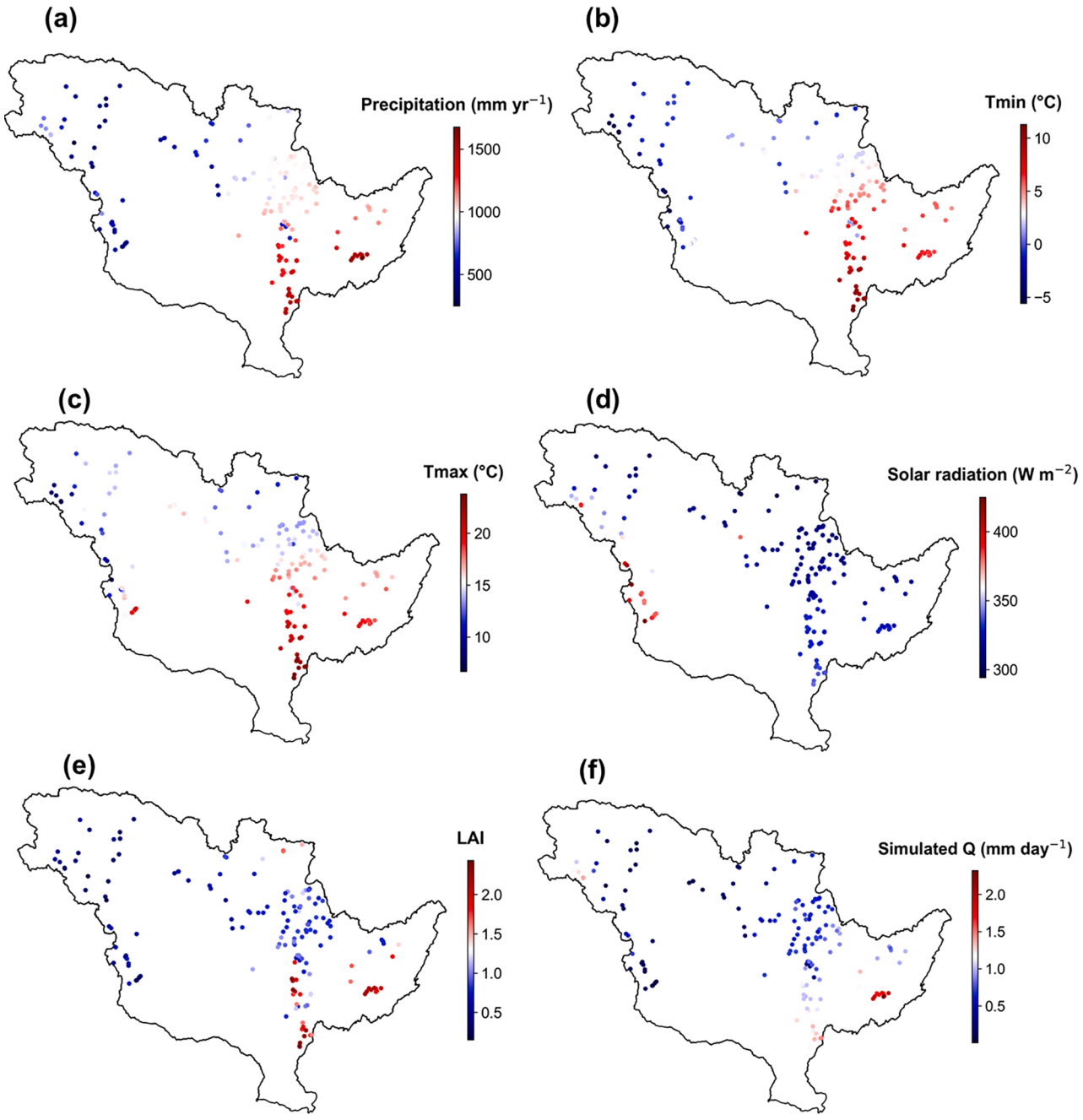

Fig. A1. Average value of dynamics attributes

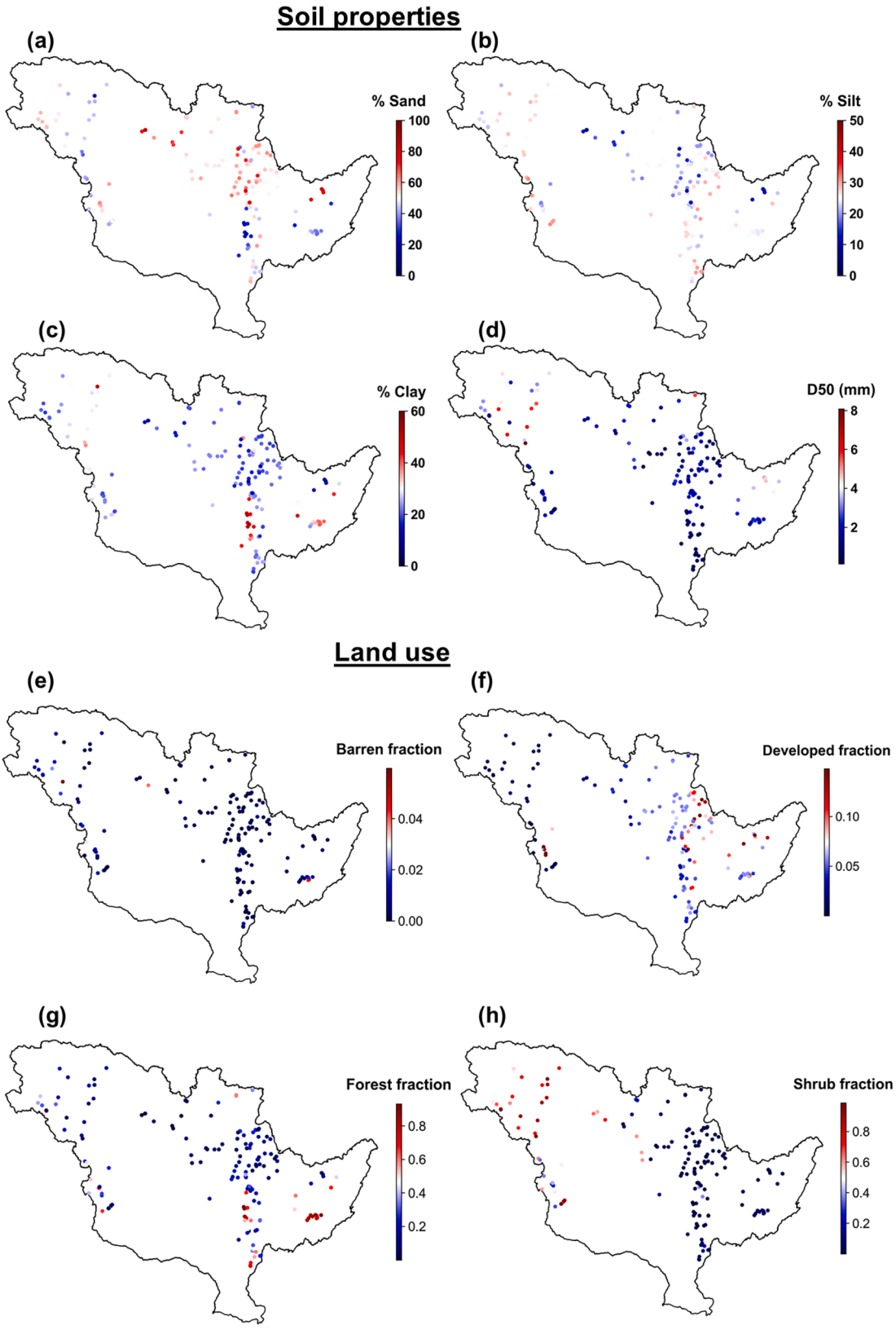

Fig. A2. Soil properties and land use of different watersheds

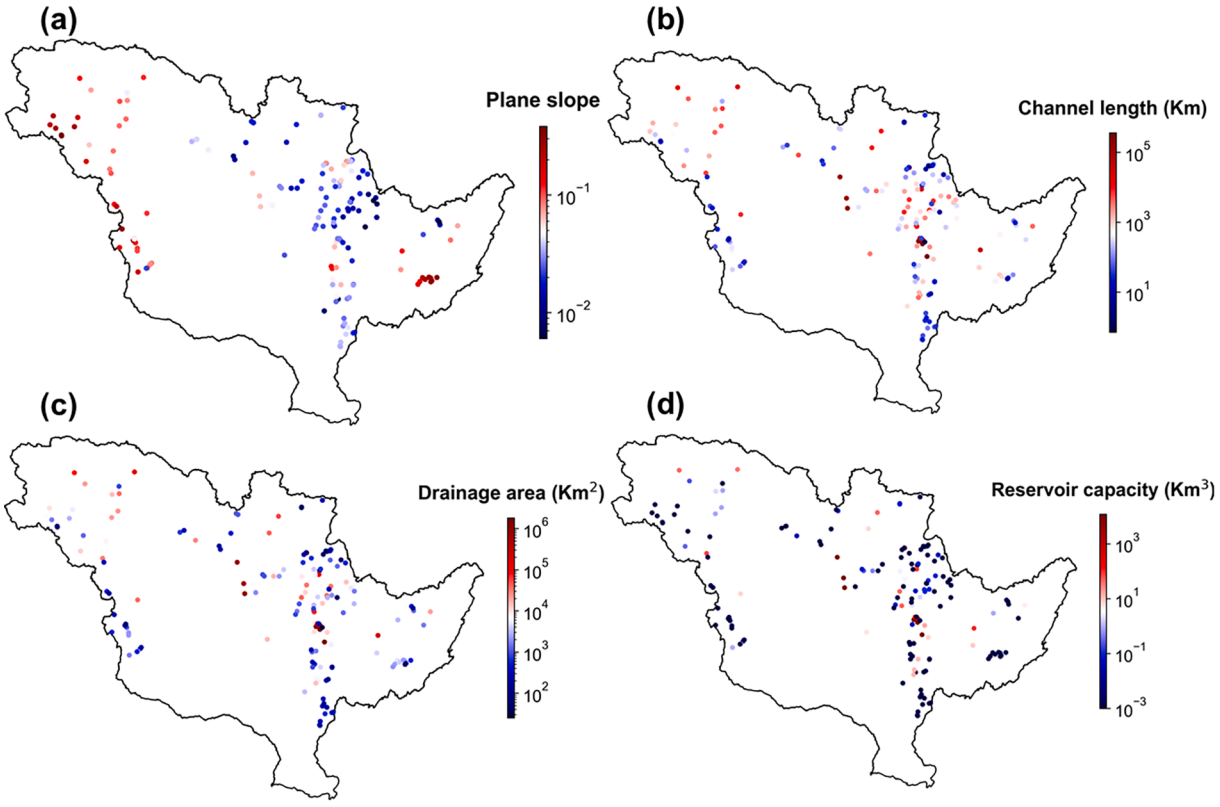

Fig. A3. Some geomorphological properties of the study watersheds

Appendix B. Optimal hyperparameters for LSTM model development

Figs. B1- shows the performance of different LSTM models with different hyperparameters used in this study.

Table B1
Range of LSTM hyperparameters.

Hyperparameter	Range
Sequence length (L)	[10, 30, 60, 120, 180]
Hidden dimension (h_{dim})	[32, 64, 128, 256]
Epochs (e)	[5, 10, 20, 30, 40]
Learning rate (α)	[0.01, 0.001, 0.0001]

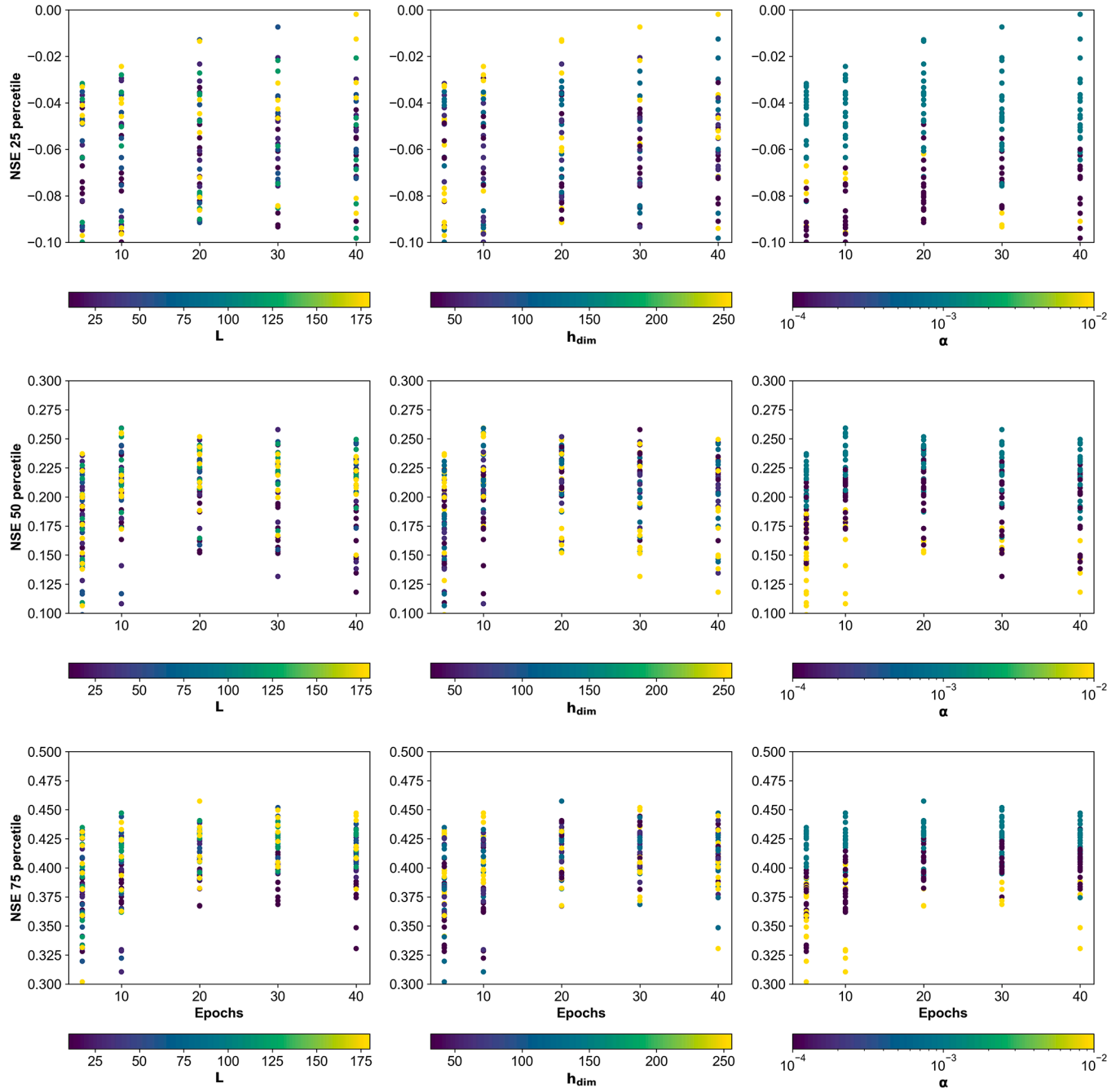

Fig. B1. Global model. Comparison of NSEs obtained by using different hyperparameters in the holdout validation scheme. The symbols h_{dim}, L and α denote hidden dimension, sequence length, and learning rate, respectively.

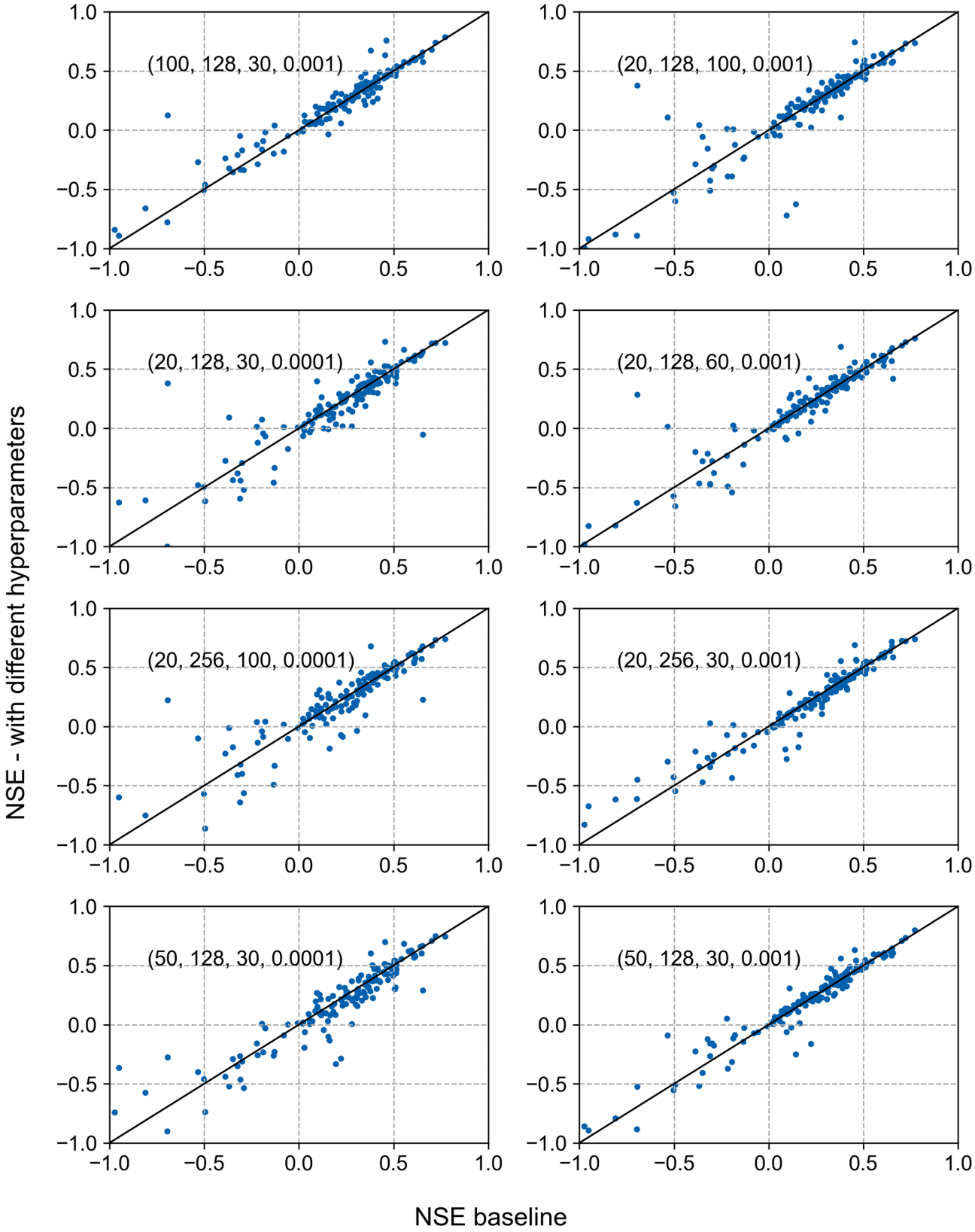

Fig. B2. Global model. Comparison of test NSEs obtained by using the LSTM models trained with different hyperparameters sets on the entire training set. The x-axis in each subplot represents the NSE obtained by using the hyperparameter set (e, h_{dim}, L, α) = (20, 128, 30, 0.001) where e, h_{dim}, L, and α stand for number of epochs, hidden dimension, sequence length, and learning rate, respectively. The figures in the bracket list the hyperparameters in order (e, h_{dim}, L, α).

Appendix C. Accuracy of VIC simulated streamflow and sediment rating curve developed using VIC simulated streamflow

Fig. C1(a) shows that the VIC simulated Q was relatively accurate in the ORB, UMRB, and LMRB. In the MB and ARW regions, the VIC simulated Q were inaccurate. Fig. C1(b) shows that the observed SSC and observed Q were more strongly correlated in the ORB, UMRB, and LMRB than in the MB region. Smaller correlations between observed Q and SSC in the MB region suggest stronger non-linearity of SSC dynamics in these watersheds.

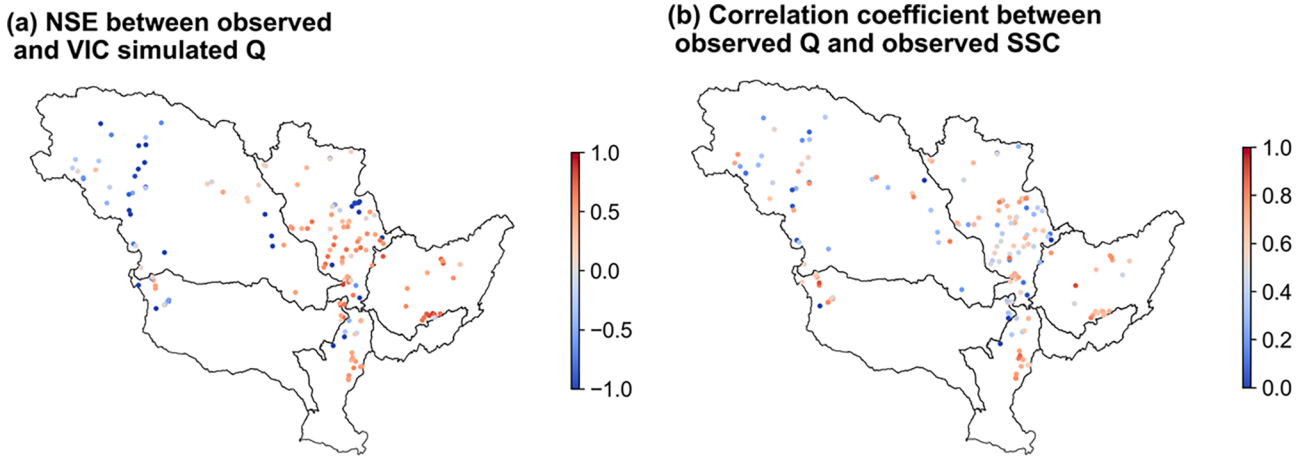

Fig. C1. (a) NSE between observed and VIC simulated streamflow (Q) and (b) correlation coefficient between observed Q and observed SSC

Appendix D. Comparison of precipitation MSWEP and DAYMET precipitation products for SSC prediction

Several gridded precipitation products are available that could be used to compute areal average precipitation. In this study, DAYMET and MSWEP products were used. DAYMET was available at 1km × 1km spatial resolution but only for North America, while MSWEP was available for the entire globe at much coarser resolution of 0.1°. Fig. D1 shows that the model performance did not change when MSWEP dataset was used instead of DAYMET, except in a few watersheds. These results suggest that even though finer resolution DAYMET data are not available for the entire globe, a global domain LSTM could still be developed using the much coarser MSWEP dataset without sacrificing the accuracy. Note that the smallest watershed in this study had total drainage area of $25km^2$ – MSWEP data may worsen the performance for watersheds much smaller than $25km^2$.

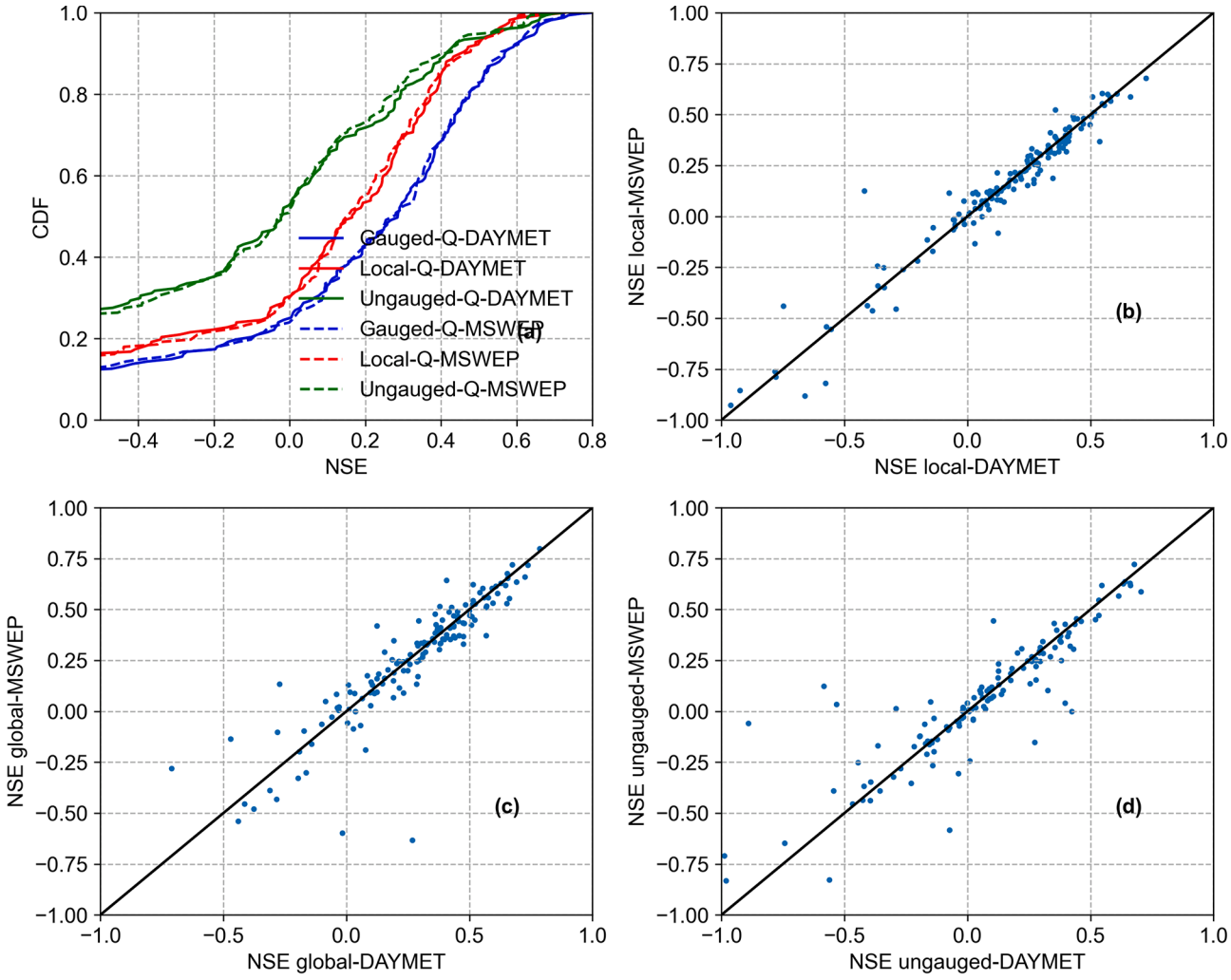

Fig. D1. Comparison of NSEs obtained by using MSWEP and DAYMET data for precipitation.

Appendix E. Impact of dams and other variables on model performance

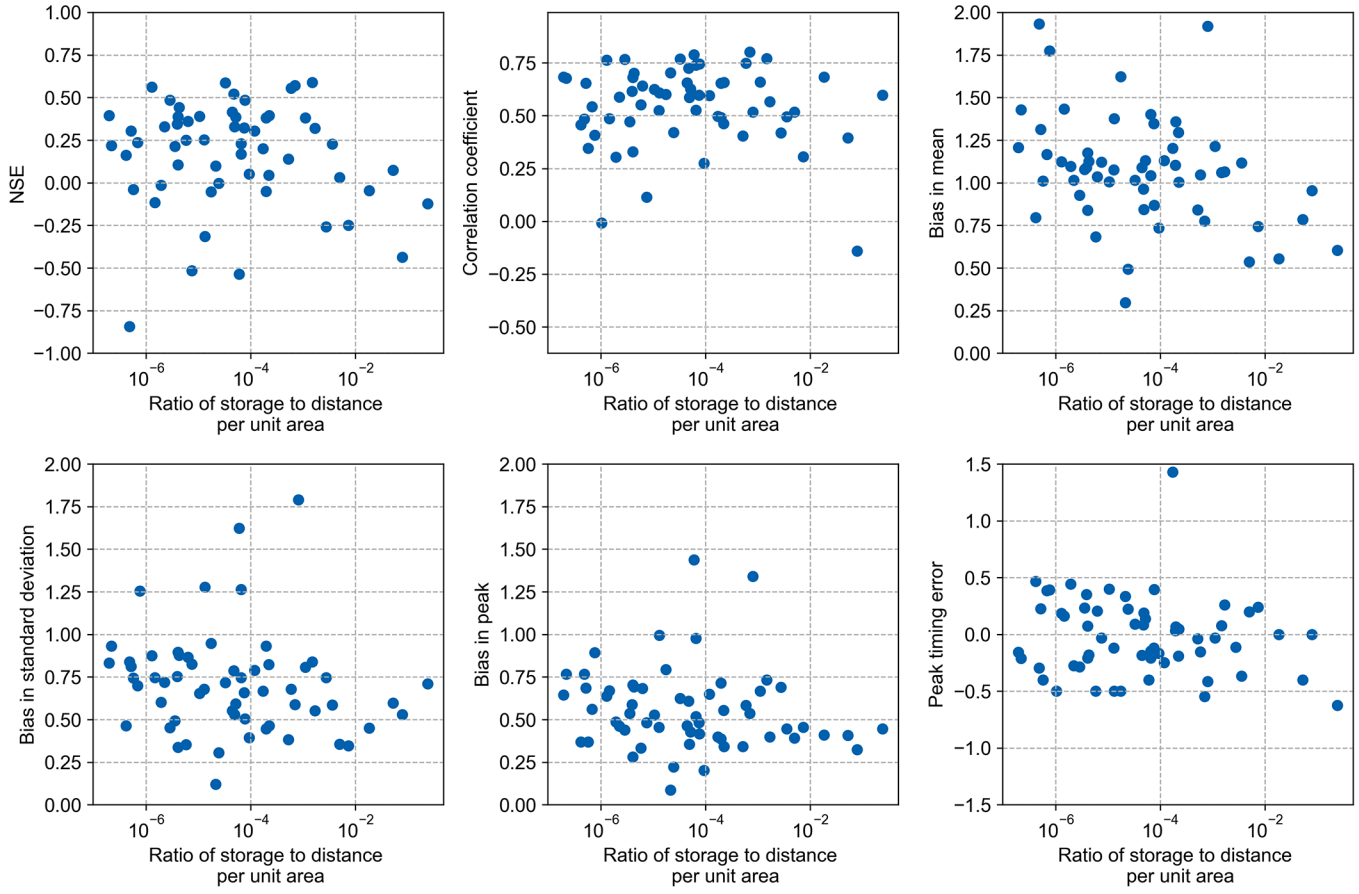

Fig. E1. Variation of performance measures with 'ratio of reservoir storage to distance per unit area' across all the watersheds. The performance measures plotted here are discussed in Equations (5)–(6). Note that the performance measures were plotted only for the watersheds where dams were present.

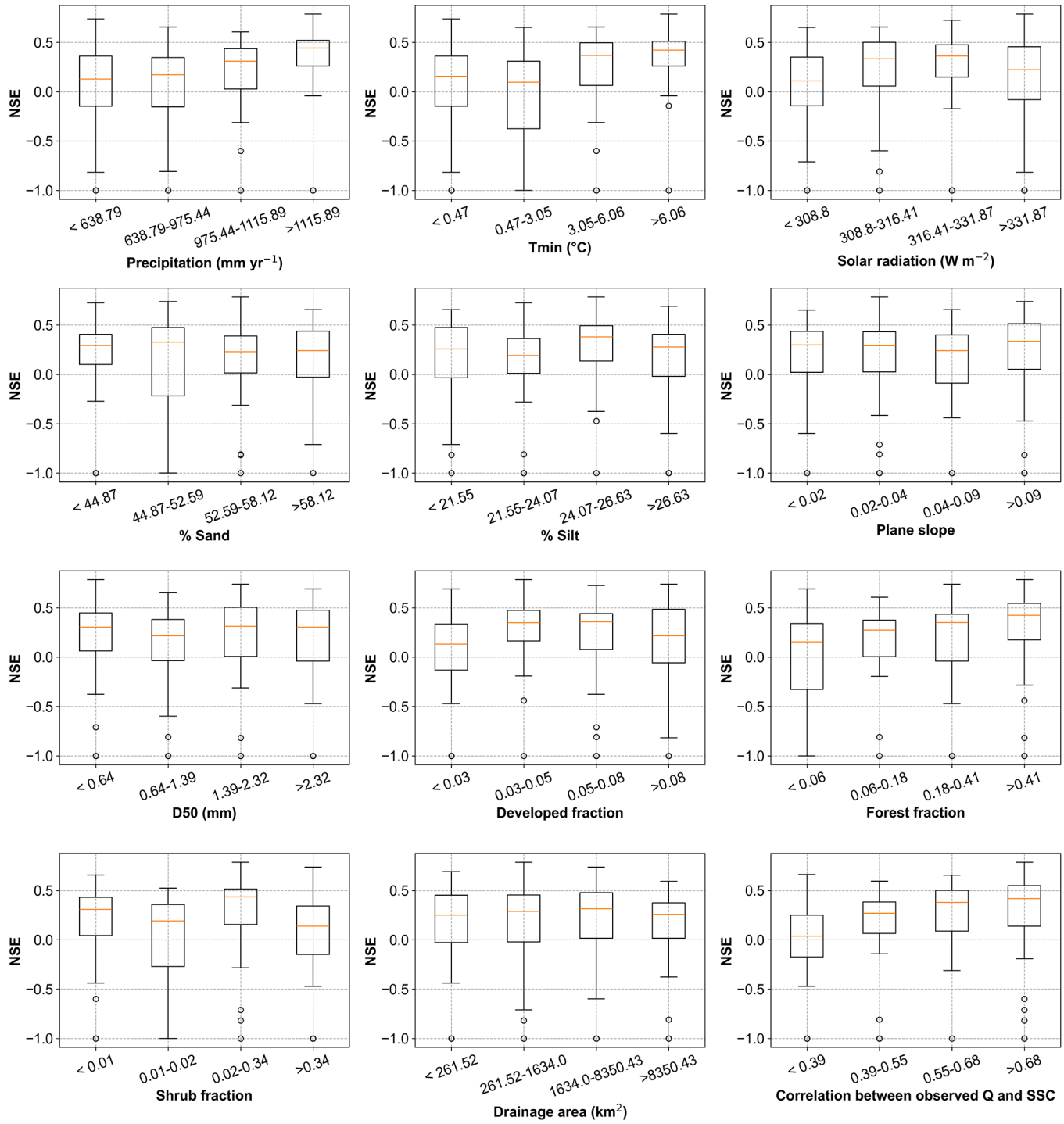

Fig. E2. Changes in test NSEs obtained by the global model (trained on NMSE objective function and simulated Q as one of the predictor variables) as the watershed attributes change

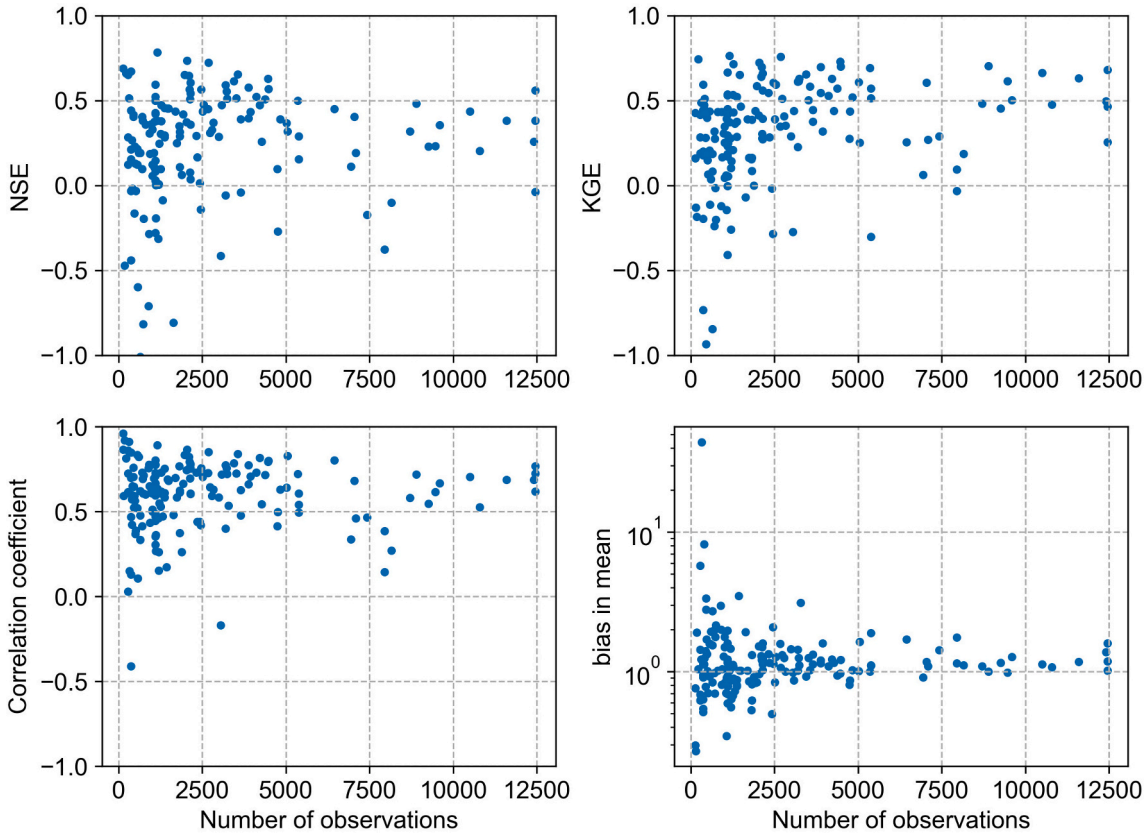

Fig. E3. Changes in model performance obtained by the global model (trained on NMSE objective function and simulated Q as one of the predictor variables) as the number of SSC observations change

Appendix F. Comparison of model performances obtained using VIC simulated streamflow and observed streamflow

Fig. F1 shows the effect of using observed streamflow on model performance. All the three models (local, global, and ungauged) performed better with observed Q. The differences between the NSEs obtained by using simulated and observed Q, however, were small for ungauged and local models. These results indicate that using a calibrated hydrological model to simulate streamflow will result in only slightly better performance compared to that using the uncalibrated VIC streamflow.

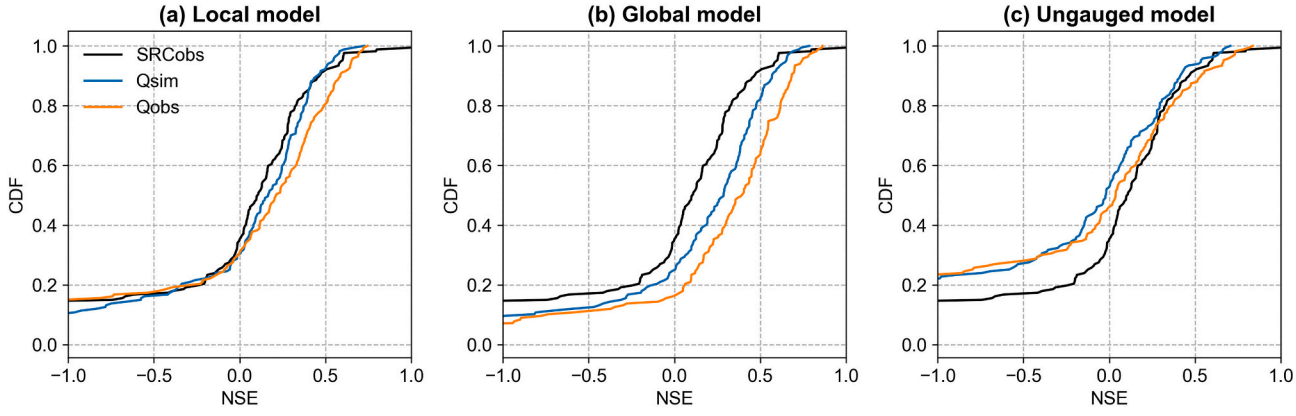

Fig. F1. Comparison of NSEs obtained by the (a) local, (b) global, and (c) ungauged models by using simulated streamflow (Qsim; blue line) and observed streamflow (Qobs; orange line). The black line shows the performance obtained by the sediment rating curve developed using the observed streamflow (SRCobs). NMSE was used as the objective function.

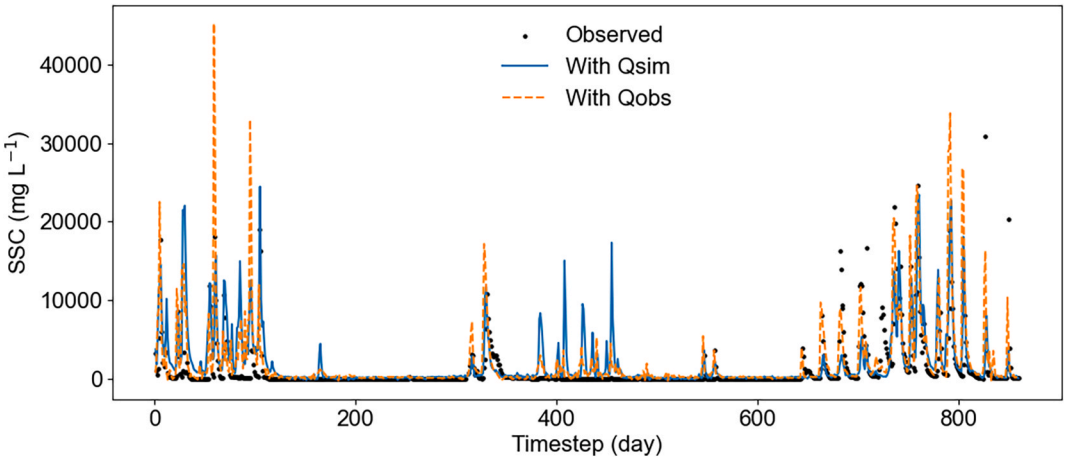

Fig. F2. Comparison of the simulated SSC obtained by the global models trained on the VIC simulated (Qsim) and observed (Qobs) streamflow. The results shown here are for the gauge USGS – 06441110 and NMSE objective function.

Data availability

A link to all the data have been provided in the manuscript.

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